DANLY IEM

PINS & BUSHINGS–METRIC



PINS & BUSHINGS – METRIC

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General Information

For replacement parts and for those customers wanting to assemble their own sets, an extensive line of catalog guide posts and bushings in both plain bearing and ball bearing styles is available for immediate delivery.

Our guide posts are available in press fit and demountable styles. Both the -82 and -83 posts can be used in a plain bearing or a ball bearing system. The NP line of guide posts fully meet NAAMS standards.

Our plain bearing bushings are available in press fit and demountable styles and are equipped with figure 8 oil grooves and lubrication fittings. They are available in three profiles: standard, short and extra long shoulder to give optimum flexibility in die set design. The bushings are also available in steel, bronze-plated and self-lubricating materials and are ideally suited for running with metric posts.

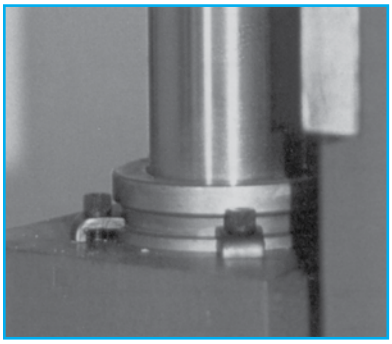
Demountable posts and bushings are tap fit into location and seat flush with the ground face of the punch holder or die shoe. They are held in place with toe clamps and screws which provide perfect alignment of the post and bushing with the bore perpendicular to the ground surface of the punch holder or shoe. The clamp and screws provide four times the holding power compared to pressed-in components, yet they can be easily removed and assembled thus simplifying die building and maintenance.

The ball bearing system includes press fit and demountable guide posts, press fit sleeves and demountable bushings as well as ball cages. The ball bearing guide posts are manufactured from hardened steel to assure free rolling of balls and high wear resistance. Each post is drilled and tapped at the bottom for mounting of the ball cage washer assembly. This unique mounting method permits the ball cage, except when under pre-load, to freely rotate 360° around the guide post thus eliminating scoring or tracking of the guide post surface. The ball bearings are arranged in the cage in a spiral pattern which also minimizes tracking or grooving and assures uniform wear.

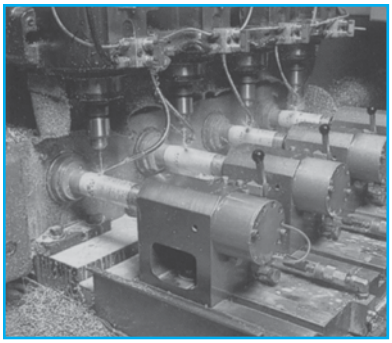
Our demountable bushings and guide posts provide ease of assembly.



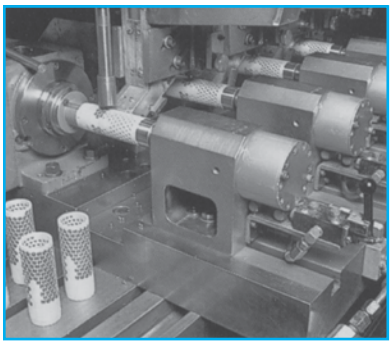
Demountable bushings are secured to the punch holder with clamps and screws, proving four times the holding power of pressed-in bushings.



Specially-designed spiral patterns are drilled into our ball cages to control tracking and grooving.



Special equipment spins ball bearings in place, then our ball cages move on to rigid quality inspection.

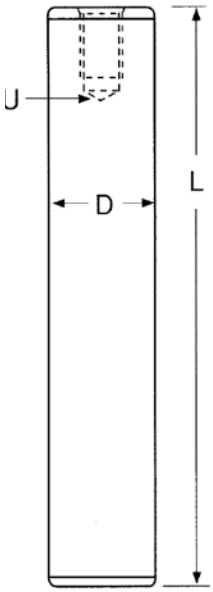


Plain & Ball Bearing
Press Fit Guide Posts



Product Features

These press fit guide posts are designed to run in both a plain and ball bearing systems. They are manufactured from high quality hardened steel and finish ground for a high precision finish. The diameters that are used in ball bearing applications are drilled and tapped on the working end for the mounting of the ball cage washer assembly. This unique mounting method permits the ball cage, except when under preload, to freely rotate 360 degrees around the guide post, thus eliminating scoring or tracking of the guide post surface.



NOTES:

- Press fit length should be equal to or greater than the diameter of the guide post.
- See pages 32–33 for die set boring specifications.
- Ball Cage washer assembly sold separately and dependent on Type I, II or III assembly methods. Refer to page 24 for washer assembly part numbers.
- The diameters that are used in ball bearing applications are drilled and tapped on the working end for the mounting of the ball cage washer assembly.

D mm	L mm	U mm	Part Number
18	100	—	5-1810-82
	110		5-1811-82
	120		5-1812-82
	130		5-1813-82
	140		5-1814-82
	150		5-1815-82
	160		5-1816-82
	170		5-1817-82
	180		5-1818-82
	190		5-1819-82
	200		5-1820-82
19	100	—	5-1910-82
	110		5-1911-82
	120		5-1912-82
	130		5-1913-82
	140		5-1914-82
	150		5-1915-82
	160		5-1916-82
	170		5-1917-82
	180		5-1918-82
	190		5-1919-82
	200		5-1920-82
24	100	—	5-2410-82
	110		5-2411-82
	120		5-2412-82
	130		5-2413-82
	140		5-2414-82
	150		5-2415-82
	160		5-2416-82
	170		5-2417-82
	180		5-2418-82
	190		5-2419-82
	200		5-2420-82
	220		5-2422-82
	240		5-2424-82
	260		5-2426-82

Plain & Ball Bearing
Press Fit Guide Posts

D mm	L mm	U mm	Part Number
25	100	M6	5-2510-82
	110		5-2511-82
	120		5-2512-82
	130		5-2513-82
	140		5-2514-82
	150		5-2515-82
	160		5-2516-82
	170		5-2517-82
	180		5-2518-82
	190		5-2519-82
	200		5-2520-82
	220		5-2522-82
	240		5-2524-82
	260		5-2526-82
30	100	—	5-3010-82
	110		5-3011-82
	120		5-3012-82
	130		5-3013-82
	140		5-3014-82
	150		5-3015-82
	160		5-3016-82
	170		5-3017-82
	180		5-3018-82
	190		5-3019-82
	200		5-3020-82
	220		5-3022-82
	240		5-3024-82
	260		5-3026-82
	280		5-3028-82
	300		5-3030-82
	320		5-3032-82
32	100	M6	5-3210-82
	110		5-3211-82
	120		5-3212-82
	130		5-3213-82
	140		5-3214-82
	150		5-3215-82
	160		5-3216-82
	170		5-3217-82
	180		5-3218-82
	190		5-3219-82
	200		5-3220-82
	220		5-3222-82
	240		5-3224-82
	260		5-3226-82
	280		5-3228-82
	300		5-3230-82
	320		5-3232-82

D mm	L mm	U mm	Part Number
38	110	—	5-3811-82
	120		5-3812-82
	130		5-3813-82
	140		5-3814-82
	150		5-3815-82
	160		5-3816-82
	170		5-3817-82
	180		5-3818-82
	190		5-3819-82
	200		5-3820-82
	220		5-3822-82
	240		5-3824-82
	260		5-3826-82
	280		5-3828-82
	300		5-3830-82
	320		5-3832-82
	360		5-3836-82
40	110	M10	5-4011-82
	120		5-4012-82
	130		5-4013-82
	140		5-4014-82
	150		5-4015-82
	160		5-4016-82
	170		5-4017-82
	180		5-4018-82
	190		5-4019-82
	200		5-4020-82
	220		5-4022-82
	240		5-4024-82
	260		5-4026-82
	280		5-4028-82
	300		5-4030-82
	320		5-4032-82
	360		5-4036-82
48	130	—	5-4813-82
	140		5-4814-82
	150		5-4815-82
	160		5-4816-82
	170		5-4817-82
	180		5-4818-82
	190		5-4819-82
	200		5-4820-82
	220		5-4822-82
	240		5-4824-82
	260		5-4826-82

D mm	L mm	U mm	Part Number
48	280	—	5-4828-82
	300		5-4830-82
	320		5-4832-82
	360		5-4836-82
	400		5-4840-82
	450		5-4845-82
50	130	M10	5-5013-82
	140		5-5014-82
	150		5-5015-82
	160		5-5016-82
	170		5-5017-82
	180		5-5018-82
	190		5-5019-82
	200		5-5020-82
	220		5-5022-82
	240		5-5024-82
	260		5-5026-82
	280		5-5028-82
	300		5-5030-82
	320		5-5032-82
	360		5-5036-82
	400		5-5040-82
	450		5-5045-82
63	200	M10	5-6320-82
	220		5-6322-82
	240		5-6324-82
	260		5-6326-82
	280		5-6328-82
	300		5-6330-82
	320		5-6332-82
	360		5-6336-82
	400		5-6340-82
	450		5-6345-82
	500		5-6350-82
80	200	M10	5-8020-82
	220		5-8022-82
	240		5-8024-82
	260		5-8026-82
	280		5-8028-82
	300		5-8030-82
	320		5-8032-82
	360		5-8036-82
	400		5-8040-82
	450		5-8045-82
	500		5-8050-82

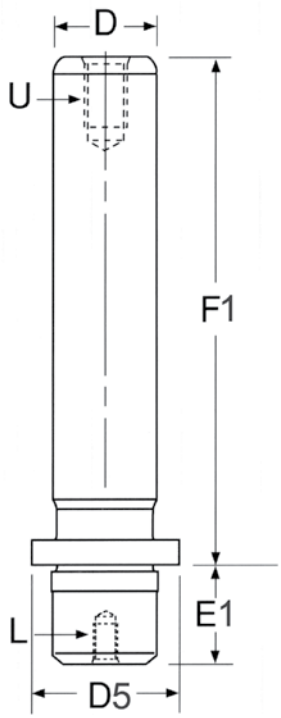
Plain & Ball Bearing Demountable Guide Posts



Product Features

These demountable guide posts are designed to run in both plain and ball bearing systems. They are manufactured from high quality hardened steel and finish ground for a high precision finish. The diameters that are used in ball bearing applications are drilled and tapped on the working end for the mounting of the ball cage washer assembly. This unique mounting method permits the ball cage, except when under preload, to freely rotate 360 degrees around the guide post, thus eliminating scoring or tracking of the guide post surface.

There are two ways of mounting the demountable posts into the die set: they can be either held in place with toe clamps and screws or they can be mounted using a retainer plug. Either mounting option offers the benefit of easy removal, even multiple times without damaging or distorting the mounting holes in the die set, thus simplifying die building and maintenance. Demountable posts are also used to replace press fit posts when the press fit hole has been damaged and the straight pin no longer fits securely in the hole.



D mm	D5 mm	E1 mm	Tap Size L	Tap Size U
18	25.6	20	M5	—
19	25.6	20	M5	—
24	32.6	24	M6	—
25	32.6	24	M6	M6
30	40.6	30	M6	—
32	40.6	30	M6	M6
38	50.8	37	M10	—
40	50.8	37	M10	M10
48	63.8	45	M12	—
50	63.8	45	M12	M10
63	76	49	M16	M10
80	93	60	M16	M10

NOTES:

- Designed to be used with bushings on pages 16-19 (not with NAAMS style bushings on pages 10-11).
- All demountable guide posts are supplied with mounting clamps and screws. See pages 34–35 for clamping dimensions or to order additional toe clamps or mounting screws.
- Ball Cage washer assembly sold separately and dependent on Type 1, 2 or 3 assembly methods. Refer to page 24 for washer assembly part numbers.
- Retainer plugs must be ordered separately. Refer to page 34 for part numbers and dimensional information.
- The diameters that are used in ball bearing applications are drilled and tapped on the working end for the mounting of the ball cage washer assembly.

Plain & Ball Bearing
Demountable Guide Posts

D mm	F1 mm	Part Number
18	80	5-1808-83
	90	5-1809-83
	100	5-1810-83
	110	5-1811-83
	120	5-1812-83
	140	5-1814-83
	160	5-1816-83
19	80	5-1908-83
	90	5-1909-83
	100	5-1910-83
	110	5-1911-83
	120	5-1912-83
	140	5-1914-83
	160	5-1916-83
24	80	5-2408-83
	90	5-2409-83
	100	5-2410-83
	110	5-2411-83
	120	5-2412-83
	130	5-2413-83
	140	5-2414-83
	150	5-2415-83
	160	5-2416-83
	170	5-2417-83
	180	5-2418-83
	190	5-2419-83
	200	5-2420-83
25	80	5-2508-83
	90	5-2509-83
	100	5-2510-83
	110	5-2511-83
	120	5-2512-83
	130	5-2513-83
	140	5-2514-83
	150	5-2515-83
	160	5-2516-83
	170	5-2517-83
	180	5-2518-83
	190	5-2519-83
	200	5-2520-83
30	100	5-3010-83
	110	5-3011-83
	120	5-3012-83
	130	5-3013-83
	140	5-3014-83
	150	5-3015-83
	160	5-3016-83
	170	5-3017-83
	180	5-3018-83
	190	5-3019-83

D mm	F1 mm	Part Number
30	200	5-3020-83
	220	5-3022-83
	240	5-3024-83
	280	5-3028-83
32	100	5-3210-83
	110	5-3211-83
	120	5-3212-83
	130	5-3213-83
	140	5-3214-83
	150	5-3215-83
	160	5-3216-83
	170	5-3217-83
	180	5-3218-83
	190	5-3219-83
	200	5-3220-83
	220	5-3222-83
	240	5-3224-83
	280	5-3228-83
38	110	5-3811-83
	120	5-3812-83
	130	5-3813-83
	140	5-3814-83
	150	5-3815-83
	160	5-3816-83
	170	5-3817-83
	180	5-3818-83
	190	5-3819-83
	200	5-3820-83
	220	5-3822-83
	240	5-3824-83
	280	5-3828-83
40	110	5-4011-83
	120	5-4012-83
	130	5-4013-83
	140	5-4014-83
	150	5-4015-83
	160	5-4016-83
	170	5-4017-83
	180	5-4018-83
	190	5-4019-83
	200	5-4020-83
	220	5-4022-83
	240	5-4024-83
	280	5-4028-83
48	110	5-4811-83
	120	5-4812-83
	130	5-4813-83
	140	5-4814-83
	150	5-4815-83
	160	5-4816-83

D mm	F1 mm	Part Number
48	170	5-4817-83
	180	5-4818-83
	190	5-4819-83
	200	5-4820-83
	220	5-4822-83
	240	5-4824-83
	280	5-4828-83
	320	5-4832-83
	360	5-4836-83
	400	5-4840-83
50	110	5-5011-83
	120	5-5012-83
	130	5-5013-83
	140	5-5014-83
	150	5-5015-83
	160	5-5016-83
	170	5-5017-83
	180	5-5018-83
	190	5-5019-83
	200	5-5020-83
	220	5-5022-83
	240	5-5024-83
	280	5-5028-83
	320	5-5032-83
	360	5-5036-83
	400	5-5040-83
63	120	5-6312-83
	140	5-6314-83
	160	5-6316-83
	180	5-6318-83
	200	5-6320-83
	220	5-6322-83
	240	5-6324-83
	280	5-6328-83
	320	5-6332-83
	360	5-6336-83
	400	5-6340-83
80	120	5-8012-83
	140	5-8014-83
	160	5-8016-83
	180	5-8018-83
	200	5-8020-83
	220	5-8022-83
	240	5-8024-83
	280	5-8028-83
	320	5-8032-83
	360	5-8036-83
	400	5-8040-83

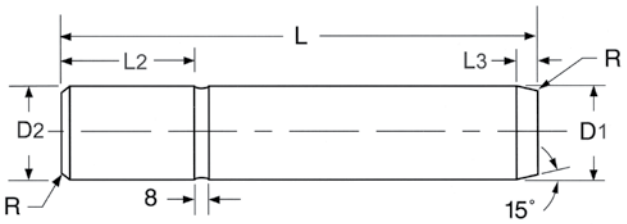
Automotive Straight Guide Posts



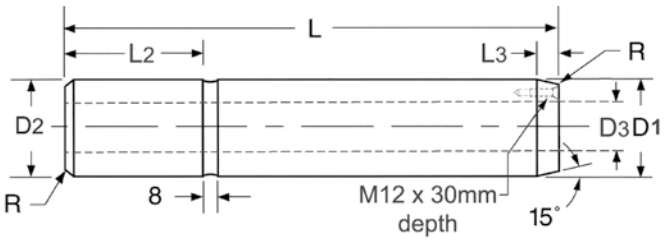
Product Features

NAAMS guide posts are manufactured in accordance with the North American Automotive Metric Standards. They are manufactured from high quality hardened steel and finish ground for a high precision finish. Hardness is RC 57-62 with a minimum depth of 0.8 min. These posts are ideally suited for the self-lubricating NAAMS bushings.

To reduce tooling weight, the 100mm and 125mm diameter posts, NPH series, are manufactured with a through hole.



For NP Series (Solid Posts)



For NPH Series (Hollow Posts)

Part Number	NAAMS Code	GM Part Number	Chrysler Part Number	Ford Part Number	D1 & D2 mm	D2 Tol r6	D3 mm	L mm	L2 mm	L3 mm	R mm	Pressed Fit Bore
NP25-80	G512508	—	19-245-6401	—	25	+.041 +.028	—	80	40	8	2	25.025 / 25.000
NP25-100	G512510	—	19-245-6402	—				100				
NP25-120	—	—	19-245-6403	—				120				
NP25-125	G512512	90.20.50-25125	19-245-6404	—				125				
NP25-140	G512514	90.20.50-25140	19-245-6405	WDX13-60-07025140				140				
NP25-160	G512516	90.20.50-25160	19-245-6406	WDX13-60-07025160				160				
NP25-180	G512518	90.20.50-25180	19-245-6407	WDX13-60-07025180				180				
NP32-100	G513210	—	19-245-6502	—	32	+.050 +.034	—	100	45	8	2	32.025 / 32.000
NP32-120	—	—	19-245-6503	—				120				
NP32-125	G513212	—	19-245-6504	—				125				
NP32-140	G513214	90.20.50-32140	19-245-6505	WDX13-60-07032140				140				
NP32-160	G513216	90.20.50-32160	19-245-6506	WDX13-60-07032160				160				
NP32-180	G513218	90.20.50-32180	19-245-6507	WDX13-60-07032180				180				
NP32-200	G513220	90.20.50-32200	19-245-6508	WDX13-60-07032200				200				
NP40-140	—	90.20.50-40140	—	—	40	+.050 +.034	—	140	56	8	2	40.025 / 40.000
NP40-160	G514016	90.20.50-40160	19-245-6606	WDX13-60-07040160				160				
NP40-180	G514018	90.20.50-40180	19-245-6607	WDX13-60-07040180				180				
NP40-200	G514020	90.20.50-40200	19-245-6608	WDX13-60-07040200				200				
NP40-225	G514022	90.20.50-40225	19-245-6609	WDX13-60-07040225				225				
NP40-250	G514025	90.20.50-40250	19-245-6610	WDX13-60-07040250				250				
NP40-280	—	90-20.50-40280	—	—				280				
NP50-160	G515016	90.20.50-50160	19-245-6806	WDX13-60-06050160	50	+.050 +.034	—	160	70	10	2.5	50.025 / 50.000
NP50-180	G515018	90.20.50-50180	19-245-6807	WDX13-60-06050180				180				
NP50-200	G515020	90.20.50-50200	19-245-6808	WDX13-60-06050200				200				
NP50-225	G515022	90.20.50-50225	19-245-6809	WDX13-60-06050225				225				
NP50-250	G515025	90.20.50-50250	19-245-6810	WDX13-60-06050250				250				
NP50-280	G515028	90.20.50-50280	19-245-6811	WDX13-60-06050280				280				
NP50-315	G515031	90.20.50-50315	19-245-6812	WDX13-60-06050315				315				
NP50-355	—	90.20.50-50355	—	—				355				

Automotive Straight Guide Posts

Part Number	NAAMS Code	GM Part Number	Chrysler Part Number	Ford Part Number	D1 & D2 mm	D2 Tol r6	D3 mm	L mm	L2 mm	L3 mm	R mm	Pressed Fit Bore
NP63-180	—	90.20.50-63180	—	—	63	+.060 +.041	—	180	80	10	2.5	63.030 / 63.000
NP63-200	G516320	90.20.50-63200	19-245-7008	WDX13-60-06063200				200				
NP63-225	G516322	90.20.50-63225	19-245-7009	WDX13-60-06063225				225				
NP63-250	G516325	90.20.50-63250	19-245-7010	WDX13-60-06063250				250				
NP63-280	G516328	90.20.50-63280	19-245-7011	WDX13-60-06063280				280				
NP63-315	G516331	90.20.50-63315	19-245-7012	WDX13-60-06063315				315				
NP63-355	G516335	90.20.50-63355	19-245-7014	WDX13-60-06063355				355				
NP63-400	G516340	90.20.50-63400	19-245-7016	WDX13-60-06063400				400				
NP63-450	—	90.20.50-63450	—	—				450				
NP63-500	G516350	90.20.50-63500	19-245-7020	—				500				
NP80-225	—	90.20.50-80225	19-245-7209	—	80	+.062 +.043	—	225	100	10	3	80.030 / 80.000
NP80-250	G518025	90.20.50-80250	19-245-7210	WDX13-60-06080250				250				
NP80-280	G518028	90.20.50-80280	19-245-7211	WDX13-60-06080280				280				
NP80-315	G518031	90.20.50-80315	19-245-7212	WDX13-60-06080315				315				
NP80-355	G518035	90.20.50-80355	19-245-7214	WDX13-60-06080355				355				
NP80-400	G518040	90.20.50-80400	19-245-7216	WDX13-60-06080400				400				
NP80-450	—	90.20.50-80450	—	WDX13-60-07080450				450				
NP80-500	G518050	90.20.50-80500	19-245-7220	WDX13-60-07080500				500				
NP100-250	—	90.20.50-100250A	—	—	100	+.073 +.051	—	250	125	10	3	100.035 / 100.000
NP100-280	—	90.20.50-100280A	—	—				280				
NP100-315	G511031	—	19-245-7412	WDX13-60-06100315				315				
NP100-355	G511035	—	19-245-7414	WDX13-60-06100355				355				
NP100-400	G511040	—	19-245-7416	WDX13-60-06100400				400				
NP100-450	—	90.20.50-100450A	19-245-7418	—				450				
NP100-500	G511050	90.20.50-100500A	19-245-7420	WDX13-60-07100500				500				
NP100-550	—	90.20.50-100550A	—	WDX13-60-07100550				550				
NP100-600	—	90.20.50-100600A	—	WDX13-60-07100600				600				
NP100-650	—	90.20.50-100650A	—	—				650				
NPH100315	—	90.20.50-100315B	—	—	100	+.073 +.051	50	315	125	10	3	100.035 / 100.000
NPH100355	—	90.20.50-100355B	—	—				355				
NPH100400	—	90.20.50-100400B	—	—				400				
NPH100450	—	90.20.50-100450B*	—	—				450				
NPH100500	—	90.20.50-100500B	—	—				500				
NPH100550	—	90.20.50-100550B*	—	—				550				
NPH100600	—	90.20.50-100600B*	—	—				600				
NP115400	G511140	90.20.50-115400	19-245-7616	—	115	+.076 +.054	—	400	140	10	3	115.035 / 115.000
NP115500	G511150	90.20.50-115500	19-245-7620	—				500				
NP125-315	—	90.20.50-125315A	—	—	125	+.088 +.063	—	315	140	10	3	125.040 / 125.000
NP125-355	—	90.20.50-125355A	—	—				355				
NP125-400	G511240	90.20.50-125400A	—	—				400				
NP125-450	G511245	90.20.50-125450A	—	—				450				
NP125-500	G511250	90.20.50-125500A	—	—				500				
NP125-550	G511255	90.20.50-125550A	—	—				550				
NP125-600	G511260	90.20.50-125600A	—	—				600				
NP125-650	G511265	90.20.50-125650A	—	—				650				
NP125-700	G511270	90.20.50-125700A	—	—				700				
NPH125315	—	90.20.50-125315B	—	—	125	+.088 +.063	65	315	140	12	3	125.040 / 125.000
NPH125355	—	90.20.50-125355B*	—	WDX13-60-05125355**				355				
NPH125400	—	90.20.50-125400B	—	WDX13-60-05125400**				400				
NPH125450	—	90.20.50-125450B	—	WDX13-60-05125450**				450				
NPH125500	—	90.20.50-125500B	—	WDX13-60-05125500**				500				
NPH125550	—	90.20.50-125550B	—	WDX13-60-05125550**				550				
NPH125600	—	90.20.50-125600B	—	WDX13-60-05125600**				600				
NPH125650	—	90.20.50-125650B	—	WDX13-60-05125650**				650				

** Not for North American Operations use.*

*** Non-standard parts. Parts must be ordered as shown.*

Guide Pins – Tap Fit

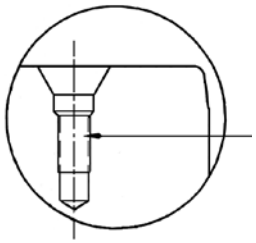
Product Features

A wide variety of pins meets most tool room requirements.

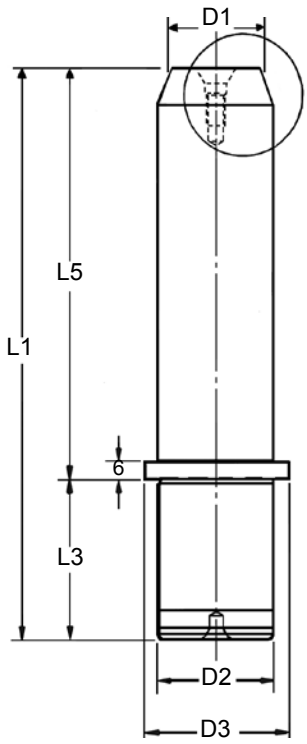
Applications:

- Transfer Presses
- Repairs

AUTOMOTIVE METRIC NAAMS STANDARD



32, 40, 50 & 63 Dia. guide posts have M8 x 1.25 -12 thread; 80, 100, 115 & 125 Dia. guide posts have M12 x 1.75 - 18 thread to facilitate handling from deep freeze.



Must be held in place with toe clamps. G720000C toe clamps & screws are included.

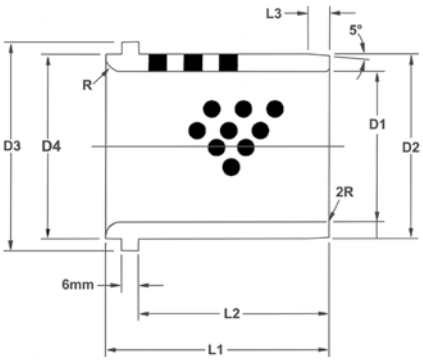
Guide Pins – with Shoulders

Part Number	D1 mm	D1 TOL g6 mm	D2 mm	D2 TOL r6 mm	D3 mm	L1 mm	L3 mm	L5 mm
G503214	32	- .009 - .025	32	+ .050 + .034	40	140	45	95
G503216						160		115
G503218						180		135
G503220						200		155
G504016	40	- .009 - .025	40	+ .050 + .034	50	160	56	104
G504018						180		124
G504020						200		144
G504022						225		169
G504025						250		194
G505016	50	- .009 - .025	50	+ .050 + .034	63	160	70	90
G505018						180		110
G505020						200		130
G505022						225		155
G505025						250		180
G505028						280		210
G505031						315		245
G506320	63	- .010 - .029	63	+ .060 + .041	80	200	80	120
G506322						225		145
G506325						250		170
G506328						280		200
G506331						315		235
G506335						355		275
G506340						400		320
G508025	80	- .010 - .029	80	+ .062 + .043	100	250	100	150
G508028						280		180
G508031						315		215
G508035						355		255
G508040						400		300
G501031	100	- .012 - .034	100	+ .073 + .051	125	315	125	190
G501035						355		230
G501040						400		275

NOTES

Automotive Demountable Self-Lubricating Guide Post Bushings





Product Features

Demountable NAAMS bushings are self-lubricating and are available in guide post and pad styles. These bushings fully conform to the NAAMS, GM, Chrysler and Ford standards.

Self-lubricating bushings contain graphite plugs which are impregnated with oil. When the bushings reach 80-90°F as a result of friction between the bushing and guide post, oil is drawn from the plug, thus lubricating the wear surface. A dark smear pattern is created on the wear surface as the oil and graphite are imbedded into the bronze or steel grain. This provides the lubrication necessary for continuous performance of the tool.

Demountable bushings are tap fit into location and seat flush with the ground face of the punch holder. The bushings are held in place with toe clamps and screws which provide perfect alignment of the bushing with the bore perpendicular to the ground surface of the punch holder. Multipe standard clamps are available.

D1 mm	D1 Tol H6	D2 mm	D2 Tol g6	D3 mm	D4 mm	L1 mm	L2 mm	L3 mm	R mm	Part Number	NAAMS Bushing Number	GM Part Number	Chrysler Part Number	Ford Part Number
25	+.013/+.000	32	-.009/-.025	40	32	40	30	4	3	NM25	G612540	90.20.55-25	19-029-1010	WDX13-60-08025
32	+.016/+.000	40	-.009/-.025	50	40	50	40	4	3	NM32	G613250	90.20.55-32	19-029-1011	WDX13-60-08032
40	+.016/+.000	50	-.009/-.025	63	50	63	50	5	3	NM40	G614063	90.20.55-40	19-029-1012	WDX13-60-08040
50	+.016/+.000	63	-.010/-.029	71	63	71	56	6	5	NM50	G615071	90.20.55-50	19-029-1013	WDX13-60-08050
63	+.019/+.000	80	-.010/-.029	90	80	80	63	8	6	NM63	G616380	90.20.55-63	19-029-1014	WDX13-60-08063
80	+.019/+.000	100	-.012/-.034	112	100	100	80	10	8	NM80	G618010	90.20.55-80	19-029-1015	WDX13-60-08080
100	+.022/+.000	125	-.014/-.039	140	125	125	106	12	10	NM100	G611012	90.20.55-100	19-029-1016	WDX13-60-08100
100	+.022/+.000	125	-.014/-.039	140	125	77	58	12	12	NM100S	—	90.20.55-100S	—	—
115	+.022/+.000	140	-.014/-.039	155	140	140	120	12	10	NM115	G611114	—	19-029-1017	—
125	+.025/+.000	160	-.014/-.039	180	160	160	132	12	12	NM125	G611216	90.20.55-125	19-029-1018	WDX13-60-08125

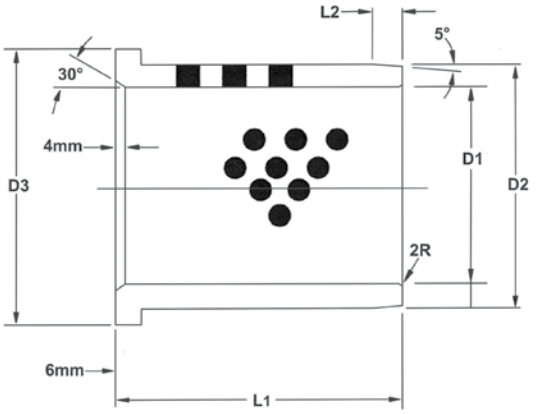
CLAMP NOTES:

- See pages 12–14 for clamp part numbers and dimensional information.
- **OUR** bushings will be supplied with (3) M8 toe clamps and screws (#6-99-1).
- **NAAMS** bushings will be supplied with (3) M8 toe clamps and screws (#6-99-1).
- **GM** bushings are supplied with NO clamps and screws. Order separately: for 25-50mm, order (1) 90.20.60A clamp per bushing; for 63-125mm, order (2) 90.20.60B clamps per bushing.
- **CHRYSLER** guide post bushings are supplied with NO clamps or screws. Order STOP/BUSHING CLAMP COMBINATION BLOCK separately. See Table 1 for sizes and part numbers.
- **Ford** bushings: 32-50mm are supplied with (1) toe clamp (#MMC0219); 63mm-100mm are supplied with (2) toe clamps (#MMC0219).

TABLE 1
STOP BLOCK/BUSHING CLAMPS

Chrysler Part Number	Bushing ID mm
19-010-1105	50mm
19-010-1106	63mm
19-010-1108	80mm
19-010-1110	100mm

Automotive Demountable Self-Lubricating PAD Bushings





D1	D1 Tol C9	D2	D2 Tol g6	D3	L1	L2	Part Number	NAAMS Code Number	GM Part Number	Chrysler Part Number	Ford Part Number
25	+.162/+.110	32	-.009/-.025	40	40	4	NM25PAD	G712540	90.30.10-25	19-029-0102	WDX13-80-09025
32	+.182/+.120	40	-.009/-.025	50	50	4	NM32PAD	G713250	90.30.10-32	19-029-0103	WDX13-80-09032
40	+.182/+.120	50	-.009/-.025	63	55	5	NM40PAD	G714055	90.30.10-40	19-029-0104	WDX13-80-09040
50	+.192/+.130	63	-.010/-.029	71	63	6	NM50PAD	G715063	90.30.10-50	19-029-0105*	WDX13-80-09050
63	+.214/+.140	80	-.010/-.029	90	75	8	NM63PAD	G716375	90.30.10-63	19-029-0106*	WDX13-80-09063
63	+.214/+.140	80	-.010/-.029	90	125	8	NM63-125PAD	—	—	19-028-0107*	—
80	+.224/+.150	100	-.012/-.034	112	90	10	NM80PAD	G718090	90.30.10-80	19-029-0108*	WDX13-80-09080
80	+.224/+.150	100	-.012/-.034	112	140	10	NM80-140PAD	—	—	19-029-0109*	—
100	+.257/+.170	125	-.014/-.039	140	115	12	NM100PAD	G711011	90.30.10-100	19-029-0110	WDX13-80-09100
100	+.257/+.170	125	-.014/-.039	140	165	12	NM100-165PAD	—	—	19-029-0111	—
125	+.300/+.200	160	-.014/-.039	180	138	12	NM125PAD	G711213	90.30.10-125	19-029-0112	—

CLAMP NOTES:

- See pages 12–14 for clamp part numbers and dimensional information.
- **OUR** PAD bushings will be supplied with (3) M8 toe clamps and screws (#6-99-1).
- **NAAMS** PAD bushings will be supplied with (3) M8 toe clamps and screws (#6-99-1).
- **GM** PAD bushings are supplied with NO clamps and screws. Order separately; for 25-50mm, order (1) 90.20.60A clamp per bushing; for 63-125mm, order (2) 90.20.60B clamps per bushing.
- **CHRYSLER** PAD bushings are supplied with (3) M10 toe clamps and screws (6-990-1). Chrysler PAD bushings with "*" DO NOT come with M10 clamps and screws (10/15/07 Die Standards). They use the PAD Bushing Retainer Block shown in Table 2 and must be ordered separately.
- **Ford** PAD bushings: 32-50mm are supplied with (1) toe clamp (#MMC0219); 63mm-100mm are supplied with (2) toe clamps (#MMC0219).

TABLE 2
PAD BUSHING RETAINER BLOCK CLAMPS

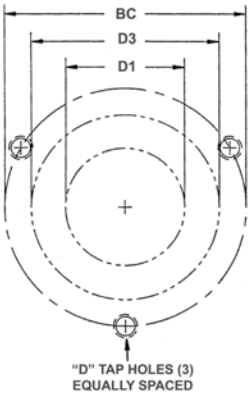
Chrysler Part Number	Pad Bushing ID mm
19-010-1185	50mm
19-010-1195[1]	50mm
19-010-1186	63mm
19-010-1196[1]	63mm
19-010-1188	80mm
19-010-1198[1]	80mm

RECOMENDED BORE SIZES FOR NAAMS GUIDE POST BUSHINGS AND PAD BUSHINGS

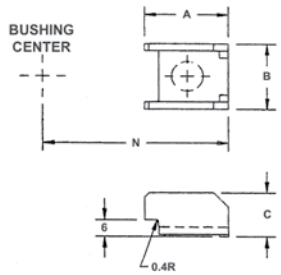
D1 mm	Wring Fit Bore
25	32.025 / 32.000
32	40.025 / 40.000
40	50.025 / 50.000
50	63.030 / 63.000
63	80.030 / 80.000
80	100.035 / 100.000
100	125.040 / 125.000
115	140.040 / 140.000
125	160.040 / 160.000

Automotive Demountable Self-Lubricating Bushing Clamps

CLAMP SPECIFICATIONS – DANLY / IEM / NAAMS / CHRYSLER



NOTE: *D1 = ID of bushing*
D3 = OD of bushing shoulder



Part Number	NAAMS Number	Chrysler Number	A mm	B mm	C mm	D mm
6-99-1	G720000C	N/A	24.6	18.9	13	M8 x 1.25
6-990-1	G730000C	19-010-0101	27.9	23.5	15.5	M10 x 1.5

PART #6-99-1 (NAAMS #G720000C)

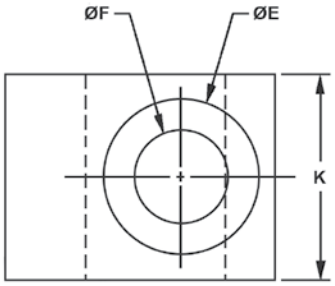
D1 mm	D3 mm	BC mm	N mm
32	50	68	45.1
40	63	81	51.6
50	71	89	55.6
63	90	108	65.1
80	112	130	76.1
100	140	158	90.1
115	155	173	97.6

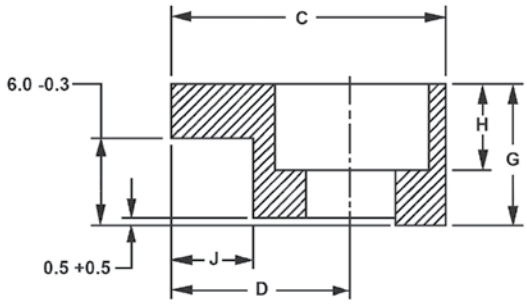
PART #6-990-1 (NAAMS #G730000C)

D1 mm	D3 mm	BC mm	N mm
32	50	71	48.4
40	63	84	54.9
50	71	92	58.9
63	90	111	68.4
80	112	133	79.4
100	140	161	93.4
115	155	176	100.9
125	180	201	113.4

CLAMP SPECIFICATIONS – GM

GM Number	Bushing I.D.	C mm	D mm	E mm	F mm	G mm	H mm	J mm	K mm	Clamps Required
90.20.60A	25	25	15	15	9	12	8.5	5	20	1
	32									
	40									
	50									
90.20.60B	63	32	21	18	11.5	16	11.5	10	25	2
	80									
	100									
	125									

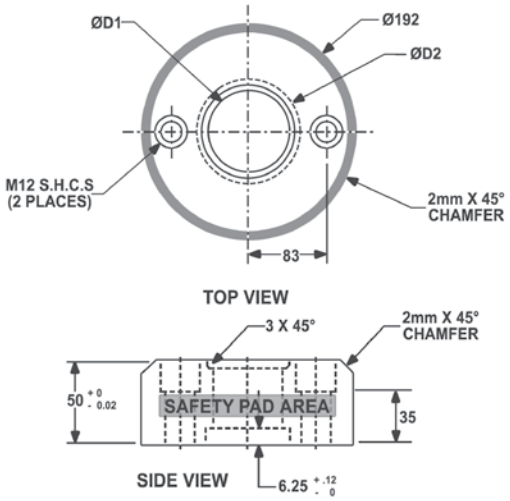




NOTE: *Clamps are to be ordered with each bushing.*

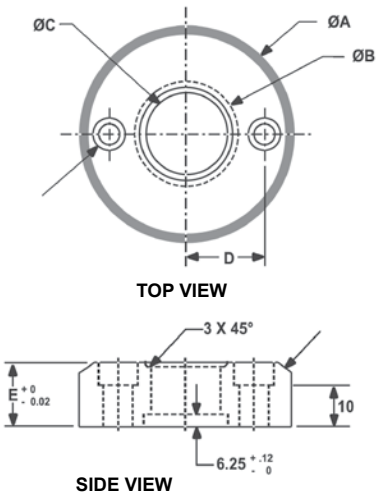
Automotive Demountable Self-Lubricating Bushing Clamps

CLAMP SPECIFICATIONS – CHRYSLER – STOP/BUSHING CLAMP COMBINATION BLOCK



D1 mm	D2 mm	Chrysler Code	Bushing I.D.
65	73	19-010-1105	50
82	92	19-010-1106	63
102	114	19-010-1108	80
127	142	19-010-1110	100

CLAMP SPECIFICATIONS – CHRYSLER – PAD BUSHING RETAINER BLOCK

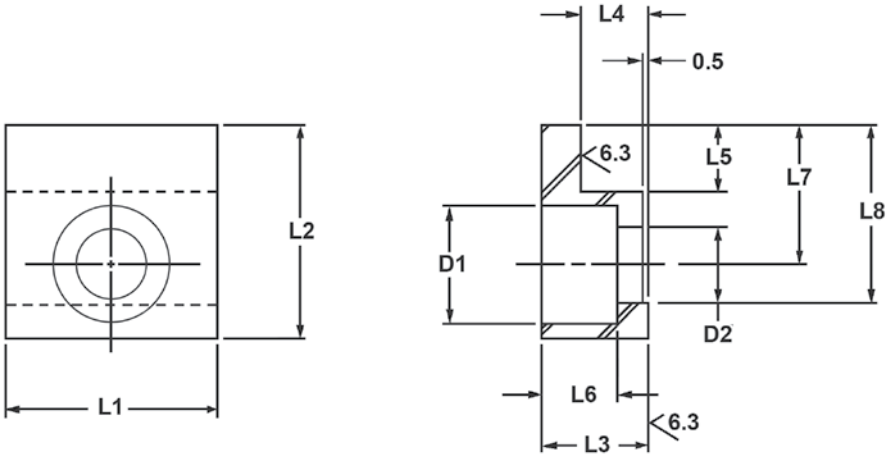


Chrysler Number	Bushing I.D.	A mm	B mm	C mm	D mm	E mm
19-010-1185	50	127	73	57	48	20
19-010-1195*	50	127	73	57	48	25
19-010-1186	63	140	92	70	57	20
19-010-1196*	63	140	92	70	57	25
19-010-1188	80	165	114	87	67	20
19-010-1198*	80	165	114	87	67	25

** For bottoming of pads/lower rings.*

Automotive Demountable Self-Lubricating Bushing Clamps

CLAMP SPECIFICATIONS – FORD – TOE CLAMPS – 25mm

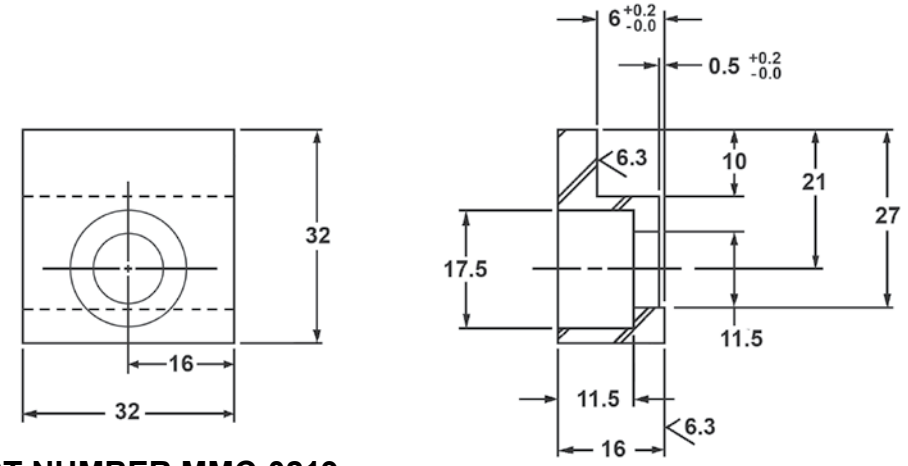


PART NUMBER MMC-0217

Ford Part Number	Bushing Size	L1 mm	L2 mm	L3 mm	L4 mm	L5 mm	L6 mm	L7 mm	L8 mm	D1 mm	D2 mm
WDX13-60-0901	25	20	20	10	6.3	5	7	12.5	16	11	7

NOTE: *For FORD 25mm bushing only*

CLAMP SPECIFICATIONS – FORD – TOE CLAMPS – 32mm – 100mm



PART NUMBER MMC-0219

Ford Part Number	Bushing Size	L1 mm	L2 mm	L3 mm	L4 mm	L5 mm	L6 mm	L7 mm	L8 mm	D1 mm	D2 mm
WDX13-60-1001	63–100	32	32	16	10	10	11.5	21	27	17.5	11.5

NOTE: *For FORD 63–100mm bushings only*

NOTES

Demountable
Plain Bearing Bushings



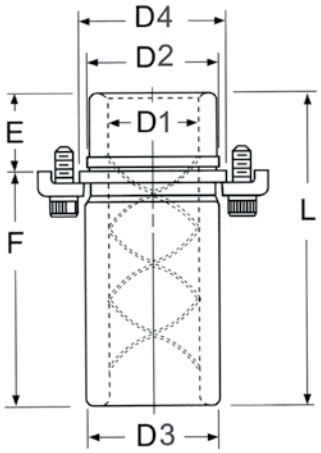
6-2438-28 & 6-2438-65

Product Features

Demountable bushings are available in three profiles: extra long, long, and short shoulder to give optimum flexibility in die set design. The bushings are manufactured from hardened steel and are ideally suited for running with press fit or demountable posts. The bronze-plated bushings offer superior resistance to seizure, the major cause of bushing wear. They are recommended in high speed applications and where high side thrust loads are present.

Demountable bushings are tap fit into location and seat flush with the ground face of the punch holder. The bushings are held in place with toe clamps and screws which provide perfect alignment of the bushing with the bore perpendicular to the ground surface of the punch holder. The clamp and screws provide four times the holding power compared to pressed-in bushings, yet they can be easily removed and assembled thus simplifying die building and maintenance.

All bushings are equipped with figure 8 oil grooves and lubrication fittings.



	D1 mm	D2 mm	D3 mm	D4 mm	E mm	F mm	L mm	Steel Part Number	Bronze Plated Part Number
EXTRA LONG SHOULDER	24	38	44	47	23	75	98	6-2438-65	6-2438-28
	25	38	44	47	23	75	98	6-2538-65	6-2538-28
	30	45	51	54	25	75	100	6-3045-65	6-3045-28
	32	45	51	54	25	75	100	6-3245-65	6-3245-28
	38	54	60	63	30	85	115	6-3854-65	6-3854-28
	40	54	60	63	30	85	115	6-4054-65	6-4054-28
	48	65	75	75	35	100	135	6-4865-65	6-4865-28
	50	65	75	75	35	100	135	6-5065-65	6-5065-28

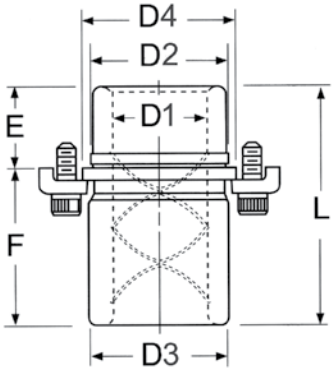
NOTES:

- All demountable bushings are supplied with mounting clamps and screws. See page 34–35 for clamping dimensions or to order additional toe clamps or mounting screws.
- Bronze-plated bushings should not be pressed-in or honed.
- See pages 32–33 for die set boring specifications.

Demountable
Plain Bearing Bushings



6-1828-64 & 6-1828-24

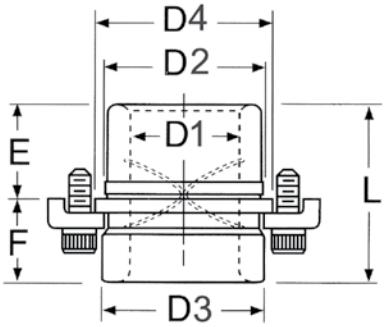


LONG SHOULDER

D1 mm	D2 mm	D3 mm	D4 mm	E mm	F mm	L mm	Steel Part Number	Bronze Plated Part Number
18	28	28	32	18	32	50	6-1828-64	6-1828-24
19	28	28	32	18	32	50	6-1928-64	6-1928-24
24	38	44	47	23	47	70	6-2438-64	6-2438-24
25	38	44	47	23	47	70	6-2538-64	6-2538-24
30	45	51	54	25	50	75	6-3045-64	6-3045-24
32	45	51	54	25	50	75	6-3245-64	6-3245-24
38	54	60	63	30	50	80	6-3854-64	6-3854-24
40	54	60	63	30	50	80	6-4054-64	6-4054-24
48	65	73	75	35	50	85	6-4865-64	6-4865-24
50	65	73	75	35	50	85	6-5065-64	6-5065-24
63	81	90	93	48	52	100	6-6381-64	6-6381-24
80	100	110	115	48	52	100	6-8010-64	6-8010-24



6-1828-63 & 6-1828-23

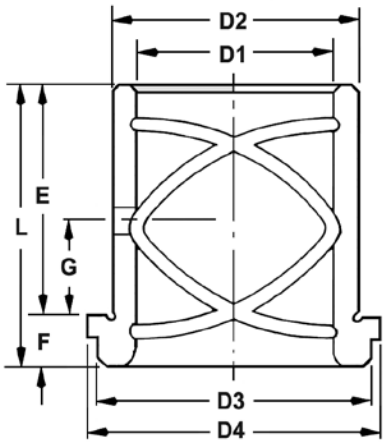


SHORT SHOULDER

D1 mm	D2 mm	D3 mm	D4 mm	E mm	F mm	L mm	Steel Part Number	Bronze Plated Part Number
18	28	28	32	18	16	34	6-1828-63	6-1828-23
19	28	28	32	18	16	34	6-1928-63	6-1928-23
24	38	44	47	23	21	44	6-2438-63	6-2438-23
25	38	44	47	23	21	44	6-2538-63	6-2538-23
30	45	51	54	25	21	46	6-3045-63	6-3045-23
32	45	51	54	25	21	46	6-3245-63	6-3245-23
38	54	60	63	30	21	51	6-3854-63	6-3854-23
40	54	60	63	30	21	51	6-4054-63	6-4054-23
48	65	73	75	35	25	60	6-4865-63	6-4865-23
50	65	73	75	35	25	60	6-5065-63	6-5065-23
63	81	90	93	48	27	75	6-6381-63	6-6381-23
80	100	110	115	48	27	75	6-8010-63	6-8010-23

Demountable Plain Bearing
Low Profile Bushings





Product Features

Low profile demountable bushings are designed so that the main body of the bushing is contained within the punch holder while only a minimum of the bushing projects below the punch holder and into the die area. With minimal bushing projection, this model is ideal for dies running in presses with automatic transfer devices. Since the bushings do not need to be removed during grinding, it is well suited for applications that require often die sharpening. The bronze-plated bushings offer superior resistance to seizure, the major cause of bushing wear. They are recommended in high speed applications and where high side thrust loads are present.

Demountable bushings are tap fit into location and seat flush with the ground face of the punch holder. The bushings are held in place with toe clamps and screws which provide perfect alignment of the bushing with the bore perpendicular to the ground surface of the punch holder. The clamp and screws provide four times the holding power compared to pressed-in bushings, yet they can be easily removed and assembled, thus simplifying die building and maintenance.

NOTES:

- All demountable bushings are supplied with mounting clamps and screws. See pages 34–35 for clamping dimensions or to order additional toe clamps or mounting screws.
- Bronze-plated bushings should not be pressed-in or honed.
- See pages 32–33 for die set boring specifications.

D1 mm	D2 mm	D3 mm	D4 mm	E mm	F mm	L mm	Hole Location G	Steel Part Number	Bronze-Plated Part Number
18	28	28	32	18	10	28	6	6-1828-68	6-1828-27
19	28	28	32	18		28		6-1928-68	6-1928-27
24	38	36	47	23		33		6-2438-68	6-2438-27
25	38	36	47	23		33		6-2538-68	6-2538-27
30	45	43	54	30		40	10	6-3045-68	6-3045-27
32	45	43	54	30		40		6-3245-68	6-3245-27
38	54	48	63	38	14	52		6-3854-68	6-3854-27
40	54	48	63	38		52		6-4054-68	6-4054-27
48	65	64	75	48		62	20	6-4865-68	6-4865-27
50	65	64	75	48		62		6-5065-68	6-5065-27
63	81	79	93	61		75		6-6381-68	6-6381-27
80	100	99	115	78		92	32	6-8010-68	6-8010-27

Self-Lubricating Ejector Bushings

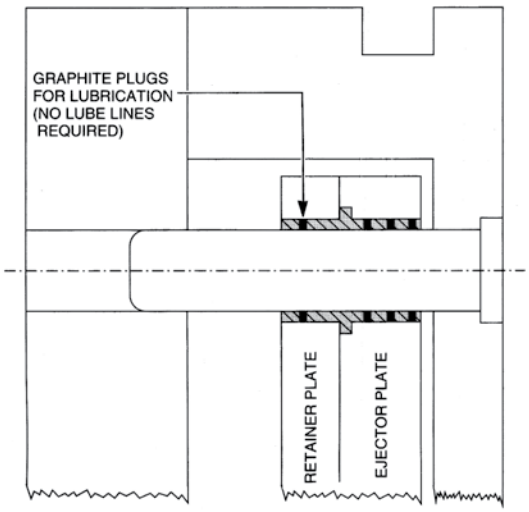
Product Features

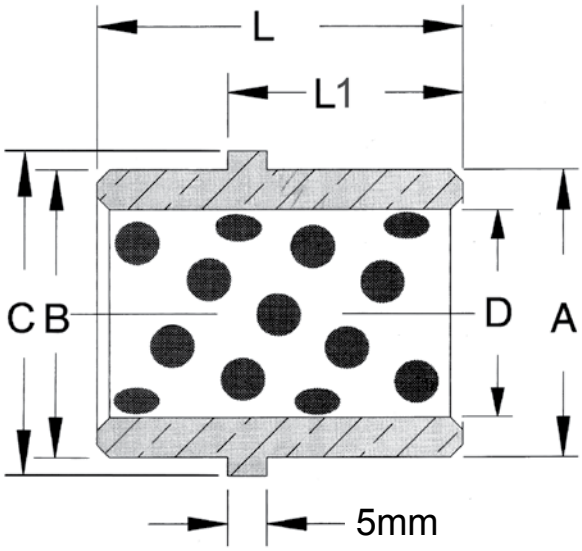
These self-lubricating bushings contain graphite plugs which are impregnated with oil. When the bushing reach 80-90°F as a result of friction between the bushing and guide post, oil is drawn from the plug, thus lubricating the wear surface. A dark smear pattern is created on the wear surface as the oil and graphite are imbedded into the bronze or steel grain. This provides the lubrication necessary for continuous performance of the tool.



NOTES:

- See pages 32–33 for die set boring specifications.





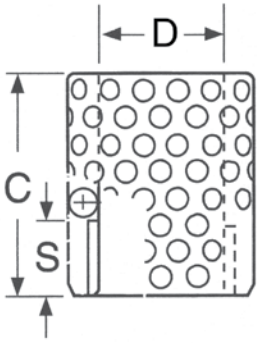
D mm	A mm	B mm	C mm	L1 mm	L mm	Part Number
20	26	26	28	25	37	MME20
25	32	32	35	29	45	MME25
30	38	38	41	29	45	MME30

Ball Bearing Cages

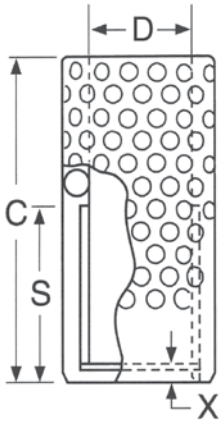


Product Features

Ball cages are manufactured from a heat treated aluminum alloy which provides tough, wear resistant qualities. The ball bearings are vacuum degassed quality, fatigue resistant steel and are inspected to ensure roundness, smoothness and dimensional conformance. The ball bearings are arranged in the cage in a spiral pattern to minimize tracking or grooving and assure uniform wear. Ball cages are mounted to drilled and tapped guide posts by a special washer assembly which permits the cage to rotate freely around the guide post when not under preload.



TYPE I
For Type I
Ball Bearing
Bushing Assembly



TYPE II
For Type II and Type III
Ball Bearing
Bushing Assemblies

D mm	C mm	S mm	X mm	Type I Part Number	Type II Part Number
25	36	11.5	4.2	6-2503-81	—
	48	17.5		6-2504-81	—
	55	31		—	6-2505-82
	70	40		—	6-2507-82
	90	47		—	6-2509-82
	100	55		—	6-2510-82
	110	65		—	6-2511-82
32	36	11.5	4.2	6-3203-81	—
	48	17.5		6-3204-81	—
	70	40		—	6-3207-82
	90	47		—	6-3209-82
	105	55		—	6-3210-82
	115	65		—	6-3211-82
	125	75		—	6-3212-82
	135	85		—	6-3213-82
40	48	17.5	5.8	6-4004-81	—
	60	23.5		6-4006-81	—
	70	40		—	6-4007-82
	85	48		—	6-4008-82
	105	56		—	6-4010-82
	115	66		—	6-4011-82
	125	76		—	6-4012-82
	135	86		—	6-4013-82
	145	96		—	6-4014-82
	155	107		—	6-4015-82

D mm	C mm	S mm	X mm	Type I Part Number	Type II Part Number
50	70	28.5	7.0	6-5007-81	—
	84	35.5		6-5008-81	—
	105	56		—	6-5010-82
	120	65		—	6-5012-82
	140	76		—	6-5014-82
	150	86		—	6-5015-82
	160	96		—	6-5016-82
	170	108		—	6-5017-82
	185	121		—	6-5018-82
	195	133		—	6-5019-82
63	98	42.4	7.0	6-6309-81	—
	145	76		—	6-6314-82
	165	86		—	6-6316-82
	180	96		—	6-6318-82
	190	107		—	6-6319-82
	205	121		—	6-6320-82
	215	132		—	6-6321-82
80	98	42.5	8.5	6-8009-81	—
	145	76		—	6-8014-82
	165	86		—	6-8016-82
	180	96		—	6-8018-82
	190	107		—	6-8019-82
	205	121		—	6-8020-82
	215	132		—	6-8021-82

Ball Bearing
Demountable Bushings

Product Features

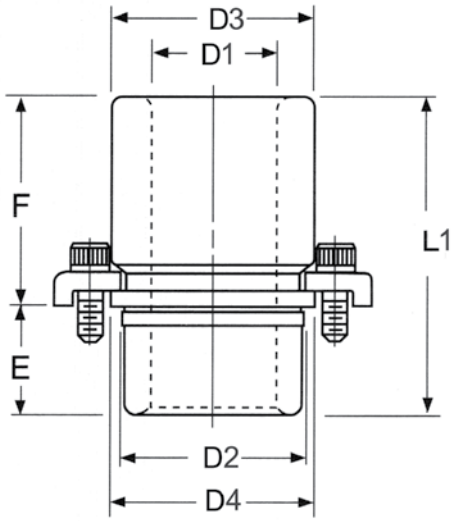
Demountable bushings are tap fit into location and seat flush with the ground face of the die holder. The bushings are held in place with toe clamps and screws which provide perfect alignment of the bushing with the bore perpendicular to the ground surface of the punch holder. The clamp and screws provide four times the holding power compared to pressed-in bushings, yet they can be easily removed and assembled thus simplifying die building and maintenance.



D mm	D1 mm	D2 mm	D3 mm	D4 mm	E mm	F mm	L1 mm	Part Number
25	33	45	50.8	55	30	35	65	6-2503-85
						50	80	6-2505-85
						65	95	6-2506-85
32	40	54	55.88	64	30	50	80	6-3205-85
						65	95	6-3206-85
						80	110	6-3208-85
40	48	65	73.03	77	30	50	80	6-4005-85
						65	95	6-4006-85
						80	110	6-4008-85
						100	130	6-4010-85
50	62	81	88.73	95	50	60	110	6-5006-85
						80	130	6-5008-85
						100	150	6-5010-85
						120	170	6-5012-85
63	75	95	103.99	109	50	100	150	6-6310-85
						120	170	6-6312-85
						140	190	6-6314-85
80	92	108	114.30	125	61	100	161	6-8010-85
						117	178	6-8011-85
						140	200	6-8014-85

NOTES:

- All demountable bushings are supplied with mounting clamps and screws. See pages 34–35 for clamping dimensions or to order additional toe clamps or mounting screws.
- See pages 32–33 for die set boring specifications.

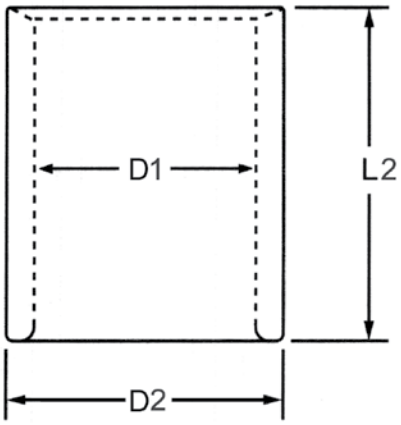


Ball Bearing
Straight Sleeve Bushings



Product Features

Press fit bushings are manufactured from high quality hardened steel, the bushings are finish ground for a press fit. Like all ball bearing components, these straight sleeve bushings are completely inter-changeable.



INSTALLATION INSTRUCTIONS

In order to avoid the bushing close-in which occurs as a result of press-fit, these bushings should be retained with a Bushing Mount. When so installed, it is not necessary to hone the bushing bore after installation, and the bushing fit will be correct.

APPLICATION OF THE BUSHING MOUNT

1. Degrease bushing OD and die shoe bore with alcohol, acetone or other volatile solvent and wipe dry.
2. Apply Bushing Mount sparingly and wring bushing into die shoe.
3. Allow a 4-hour cure at 72° F (23° C). Do not disturb bushing until cure is complete.

Bushing Mount Part number: 9-60-82

Post Diameter mm	D1 mm	D2 mm	L2 mm	Part Number
25	33	45	65	6-2506-86
			80	6-2508-86
			95	6-2509-86
			110	6-2511-86
			130	6-2513-86
32	40	54	80	6-3208-86
			95	6-3209-86
			110	6-3211-86
			130	6-3213-86
			150	6-3215-86
			170	6-3217-86
40	48	65	80	6-4008-86
			95	6-4009-86
			110	6-4011-86
			130	6-4013-86
			150	6-4015-86
			170	6-4017-86
			190	6-4019-86
			215	6-4021-86

Post Diameter mm	D1 mm	D2 mm	L2 mm	Part Number
50	62	81	110	6-5011-86
			130	6-5013-86
			150	6-5015-86
			170	6-5017-86
			190	6-5019-86
			215	6-5021-86
			240	6-5024-86
			265	6-5026-86
63	75	95	150	6-6315-86
			170	6-6317-86
			190	6-6319-86
			215	6-6321-86
			240	6-6324-86
			265	6-6326-86
80	92	112	150	6-8015-86
			170	6-8017-86
			190	6-8019-86
			215	6-8021-86
			240	6-8024-86
			265	6-8026-86

Demountable Ball Bearing Stripper Bushings and Cages

Product Features

Type III cages are made using the same material and control standards as our Type I & II cages. Type III cages use an external snap ring to act as a cage carrier and stop. This allows the cage to rotate freely around the guide post when not under preload and prevents the cage from pushing through the guide bushing at full stroke. Ball bearings are arranged in a spiral pattern in the cage to minimize tracking or grooving and assure uniform wear.

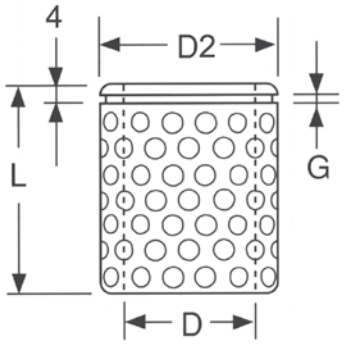
Low profile demountable bushings give maximum flexibility in die set design. The low profile bushing, coupled with low mount clamping, provide maximum available die set use. The bushings are manufactured from hardened steel and are designed for use with Type III cage components. These demountable bushings are a tap fit into location and seat flush with the ground face of the die holder. The bushings are held in place with low mount clamps and screws which provide perfect alignment of the bushing with the bore perpendicular to the ground surface of the die holder.



6-2505-83 & 6-2525-87

Cage Data

D mm	L mm	G mm	D2 mm	Part Number
25	40	1.6	30.3	6-2504-83
	52			6-2505-83
32	40	1.9	36	6-3204-83
	52			6-3205-83
40	52	1.9	45.5	6-4005-83
	64			6-4006-83
50	74	2.2	57	6-5007-83
	88			6-5008-83

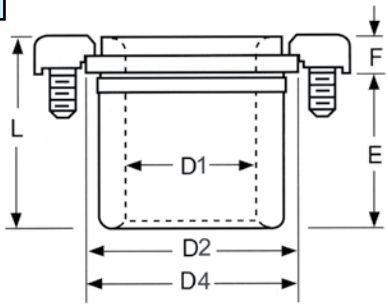


Bushing Data

Post Diameter mm	D1 mm	D2 mm	D4 mm	E mm	F mm	L mm	Part Number
25	33	45	51	20	10	30	6-2520-87
				25		35	6-2525-87
32	40	54	60	20	10	30	6-3220-87
				25		35	6-3225-87
				32		42	6-3232-87
40	48	65	75	29	10	39	6-4029-87
				36		46	6-4036-87
				44		54	6-4044-87
50	62	81	91	36	10	46	6-5036-87
				44		54	6-5044-87

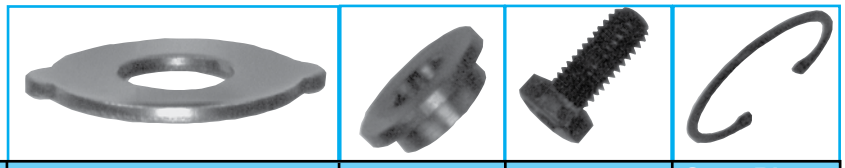
NOTES:

- All demountable bushings are supplied with mounting clamps and screws. See pages 34–35 for clamping dimensions or to order additional toe clamps or mounting screws.
- See pages 32–33 for die set boring specifications.



Mounting Accessories

BALL CAGE WASHER ASSEMBLY



Nominal Post Diameter A mm	Washer Assembly Part Number		Stop Washer-Retainer		Bushing Stop Washer	Hex Head Screw	Snap Ring-Ball Bearing Cages
	Type I Assembly	Type II Assembly	Type I	Type II and III	Type II and III	Type I, II and III	Type II and III
25	6-0025-81	6-0025-82	6-2500-85	6-2500-83	6-2500-84	6-0620-933 M6 x 20 DIN933 8G or 10K	6-2500-86
32	6-0032-81	6-0032-82	6-3200-85	6-3200-83			6-3200-86
40	6-0040-81	6-0040-82	6-4000-85	6-4000-83	6-4000-84	6-1025-933 M10 x 25 DIN933 8G or 10K	6-4000-86
50	6-0050-81	6-0050-82	6-5000-85	6-5000-83			6-5000-86
63	6-0063-81	6-0063-82	6-6300-85	6-6300-83			6-6300-86
80	6-0080-81	6-0080-82	6-8000-85	6-8000-83			6-8000-86

NOTES:

- **TYPE I** Assembly includes stopwasher retainer and head hex screw.
- **TYPE II** Assembly includes stopwasher retainer, bushing stopwasher, hex head screw and snap ring.

BUSHING MOUNT/RETAINING COMPOUND

Part Number	Volume
9-60-82	118 mL

NOTES:

- A bushing mount/retaining compound is used when installing straight sleeve ball bearing bushings to avoid bushing close-in as a result of press-fit.

BALL CAGE WASHER ASSEMBLY

Guide Post Tap Sizes		Required Seating Torque	
Diameter O mm	Tap Size	N-m	In-lbs
25 – 32	M6 x 1.0	36	320
40 – 50	M10 x 1.5	98	867
63 – 80	M10 x 1.5	236	2088

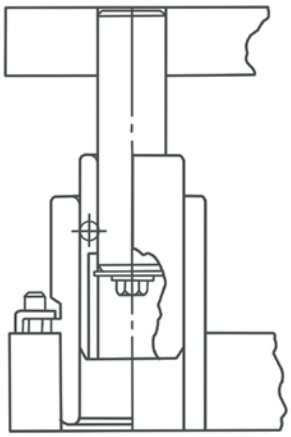
BALL BEARING BUSHING LUBRICATION RECOMMENDATION

During the operation of the Ball Bearing Bushing Assembly, add lubricant once each 8-hour shift by spray or brush application. Use a refined mineral oil of viscosity 290/340 SSU @ 100° F (38° C) containing “EP” additives and rust inhibitors.

Ball Bearing Components Selection Guide

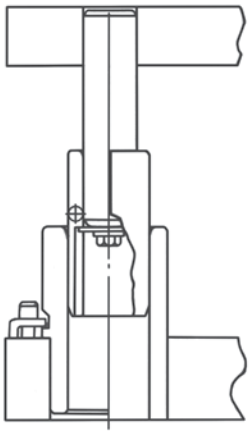
Product Features

For proper post, bushing and cage selection, the operating conditions of the die must be taken into consideration. Factors such as press speed, shut height, stroke length and operating environment all play a role in selecting the appropriate operating condition to give the best performance possible of your components. The operating conditions include full preload, relieved and total disengagement.



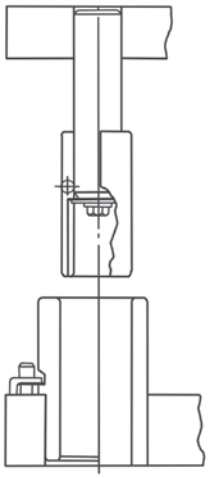
Type I
Full Pre-Load Throughout Stroke

Type I component assemblies ensure that all ball bearings remain in constant contact with the guide post and bushing throughout the entire press stroke. This assembly is recommended for higher speed, short stroke dies.



Type II
Pre-Load Relived or Partially Relieved

Type II component assemblies are designed so the ball cage never leaves the bushing; however the guide post disengages the bushing at the beginning of each stroke. This assembly is often preferred as pinch points are eliminated and foreign matter cannot get inside the assembly and damage the components. This assembly is often utilized in medium and long stroke dies.



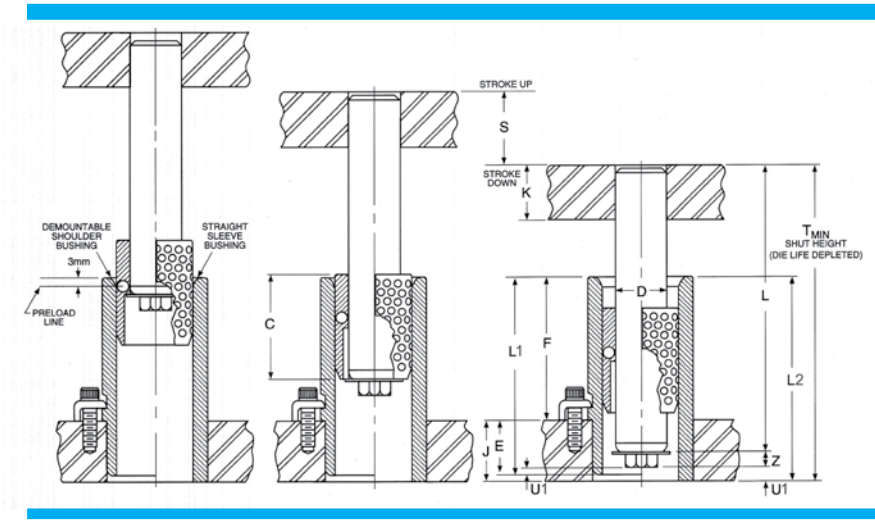
Type III
Total Disengagement of Cage from Bushing

Type III component assemblies are designed so the ball cage totally disengages from the bushing. This assembly is utilized on long stroke applications.

NOTES:

- See ball bearing lubrication recommendation on page 24.

Type I
Component Selection Guide



Post Diam. D mm	Z mm	E mm	U1 mm
25	7	30	3
32	7	30	
40	9.5	30	
50	9.5	50	
63	9.5	50	
80	9.5	50	

PRESS FIT POSTS

1. Calculate L, the desired guide post length, using one of the following 2 formulas:
   Assembly with Straight Sleeve Bushings: $L=T-U_1-Z$
   Assembly with Demountable Bushings: $L=T-U_1-Z-J+E$
2. Select a post length from the catalog that is equal to L calculated above. If the calculated L value is not a standard catalog length, you have two options. Choose the next longest length and cut off to the calculated L dimension or select a shorter length and recess the post in the punch holder to obtain correct L dimension.
   Note: *Press fit length should be equal to or greater than the diameter of the guide post.*

DEMOUNTABLE POSTS

1. Calculate F_1, the desired guide post length, using one of the following 2 formulas:
   Assembly with Straight Sleeve Bushings: $F_1=T-U_1-Z-K$ (Note $F+J+K+Y<T$)
   Assembly with Demountable Bushing: $F_1=T-U_1-Z-J+E-K$ (Note $L_2+K+Y<T$)
2. Select a post length from the catalog that is equal to F_1 calculated above. If the calculated F_1 value is not a standard catalog length, choose a catalog length that is close to but less than the calculated F_1.
   Note: *Demountable posts cannot be cut off. See page 4 for standard post lengths (F_1).*

BUSHING & CAGE SELECTION

Selection of a Type 1 Ball bearing bushing and cage assembly is based on the required stroke and the guide post diameter.

1. Determine the guide post diameter required and the stroke required.
2. Using the selection chart on page 27, find the desired stroke. Move down this column to the colored square on the horizontal line opposite the required post diameter.
3. Select the required bushing length which is listed to the left of the selected square in the columns labeled demountable shoulder and straight sleeve.
   Note: *For applications with no off-center loading, select the bushing with the shortest overall length from the selection chart. However for longer stroke applications or where side-loading may be present select the bushing with the longest possible length to provide optimal guidance.*
4. Select the required cage length which is also listed to the left of the selected square in the column labeled "Ball Cage."
   Note: *Shut height permitting, select the longest cage length possible for optimal performance.*

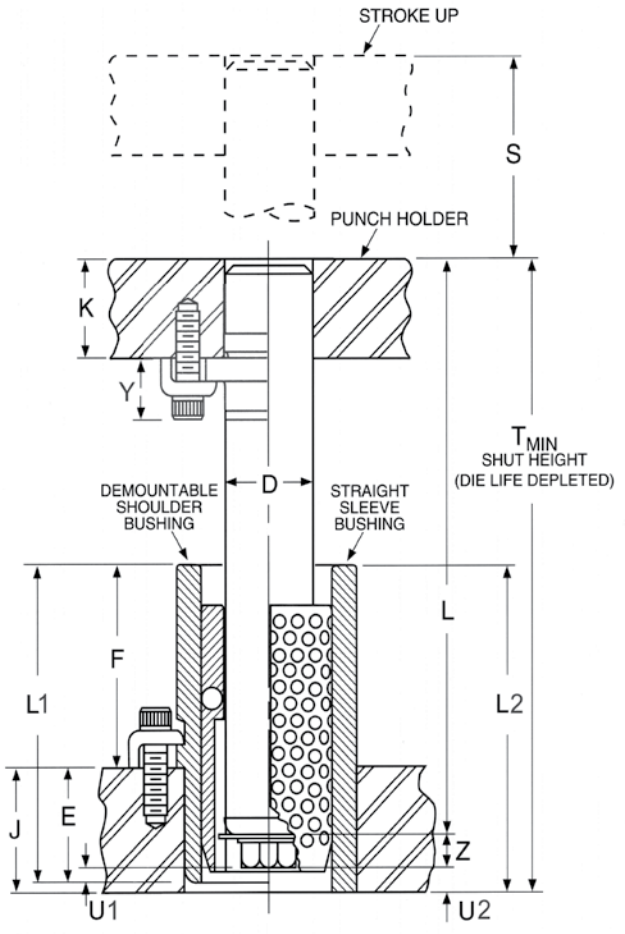
NOTES:

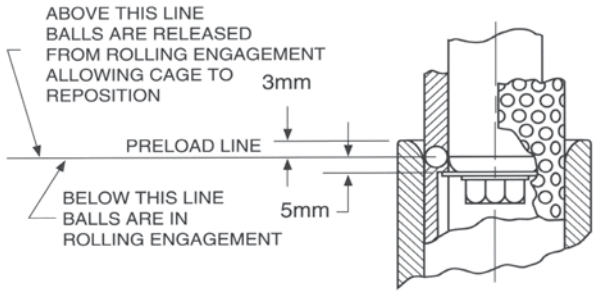
- If die grind is not required, stroke may be increased by the amount of die grind allowance, dimension X, found in the right most column of the selection chart.
- A die set designed for a particular stroke may be used in any press of lesser stroke but never in any press where the stroke is greater than originally chosen.
- Die shoe thickness must be greater than "E" dimension when shoulder bushings are selected.

Type I
Component Selection Guide

NOM. POST DIA.	BALL BUSHING			BALL CAGE	STROKE "S" INCLUDING DIE GRIND ALLOWANCE	*MAXIMUM DIE WEAR ALLOWANCE
	SHOULDER		SLEEVE			
D mm	F mm	L_1 mm	L_2 mm	C mm	20 40 60 80 100 120 140 160	X (mm)
25	35	65	65	36	If your selection falls in a white square, use Type II or Type III.	10
	50	80	80			
	65	95	95			
	–	–	110			
	–	–	130	48		
32	50	80	80	36		10
	65	95	95			
	80	110	110			
	–	–	130			
	–	–	150			
	–	–	170	48		
40	50	80	80	48		10
	65	95	95			
	80	110	110			
	100	130	130			
	–	–	150			
	–	–	170			
	–	–	190			
	–	–	215	60		
50	60	110	110	70		10
	80	130	130			
	100	150	150			
	120	170	170			
	–	–	190			
	–	–	215			
	–	–	240			
	–	–	265	84		
63	100	150	150	98		15
	120	170	170			
	140	190	190			
	–	–	215			
	–	–	240			
	–	–	265			
80	–	–	150	98		15
	–	–	170			
	–	–	190			
	–	–	215			
	–	–	240			
	–	–	265			

Type II & III
Component Selection Guide





D mm	Z mm	E mm	U1 mm	U2 mm	Y mm
25	10.5	30	1.5	3.5	14.7
32	10.5	30		3.5	18.9
40	13	30		4	18.9
50	13	50		4	18.9
63	13	50		4	18.9
80	13	50		4	18.9

Type II & III
Component Selection Guide

PRESS FIT POSTS

1. Calculate L, the desired guide post length, using one of the following 2 formulas:
   Assembly with Straight Sleeve Bushings: $L=T-U_2-Z$
   Assembly with Demountable Bushings: $L=T-U_2-Z-J+E$
2. Select a post length from the catalog that is equal to L calculated above. If the calculated L value is not a standard catalog length, you have two options. Choose the next longest length and cut off to the calculated L dimension or select a shorter length and recess the post in the punch holder to obtain correct L dimension.
   Note: *Press fit length should be equal to or greater than the diameter of the guide post.*

DEMOUNTABLE POSTS

1. Calculate F_1, the desired guide post length, using one of the following 2 formulas:
   Assembly with Straight Sleeve Bushings: $F_1=T-U_2-Z-K$ (Note $F+J+K+Y<T$)
   Assembly with Demountable Bushings: $F_1=T-U_1-Z-J+E-K$ (Note $L_2+K+Y<T$)
2. Select a post length from the catalog that is equal to F_1 calculated above. If the calculated F_1 value is not a standard catalog length, choose a catalog length that is close to but less than the calculated F_1.
   Note: *Demountable posts cannot be cut off. See page 4 for standard post lengths (F_1).*

BUSHING & CAGE SELECTION

Selection of a Type II and Type III Ball bearing bushing and cage assembly is based on the required stroke and the guide post diameter.

1. Determine the guide post diameter required and the stroke required.
2. Determine the desired operating condition or the extent to which the cage leaves the bushing.
3. Determine if a demountable or straight sleeve bushing is to be used.
4. Using the selection chart on pages 30 & 31, find the desired stroke (S). Move down this column to the colored square on the horizontal line opposite the required post diameter. Find the colored square in the desired operating condition.
5. Select the required bushing length which is listed to the left of the selected square in the columns labeled demountable shoulder or straight sleeve.
   Note: *For applications with no off-center loading, select the bushing with the shortest overall length from the selection chart. However for longer stroke applications or where side-loading may be present select the bushing with the longest possible length to provide optimal guidance.*
6. Select the required cage length which is also listed to the left of the selected square in the column labeled “Ball Cage.”
   Note: *Shut height permitting, select the longest cage length possible for optimal performance.*

NOTES:

- If stroking rate is under 150 rpm, Figure B (on page 28) is recommended, which allows the ball cage to reposition at each stroke.
- A die set designed for a particular stroke may be used in any press of lesser stroke but never in any press where the stroke is greater than originally chosen.
- Die shoe thickness must be greater than “E” dimension when shoulder bushings are selected.

Type II & III
Bushing & Ball Cage Selection Guide

CAUTION

Be sure bushing does not strike punch holder at minimum shut height. If this condition exists, use shorter bushing and corresponding ball cage.

Type II and III components provide Type I operating conditions

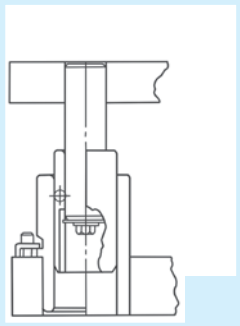


Figure A
Partial Preload

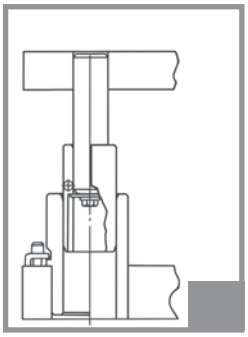


Figure B
Preload Relieved

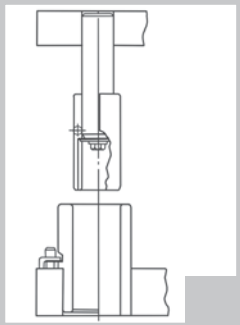


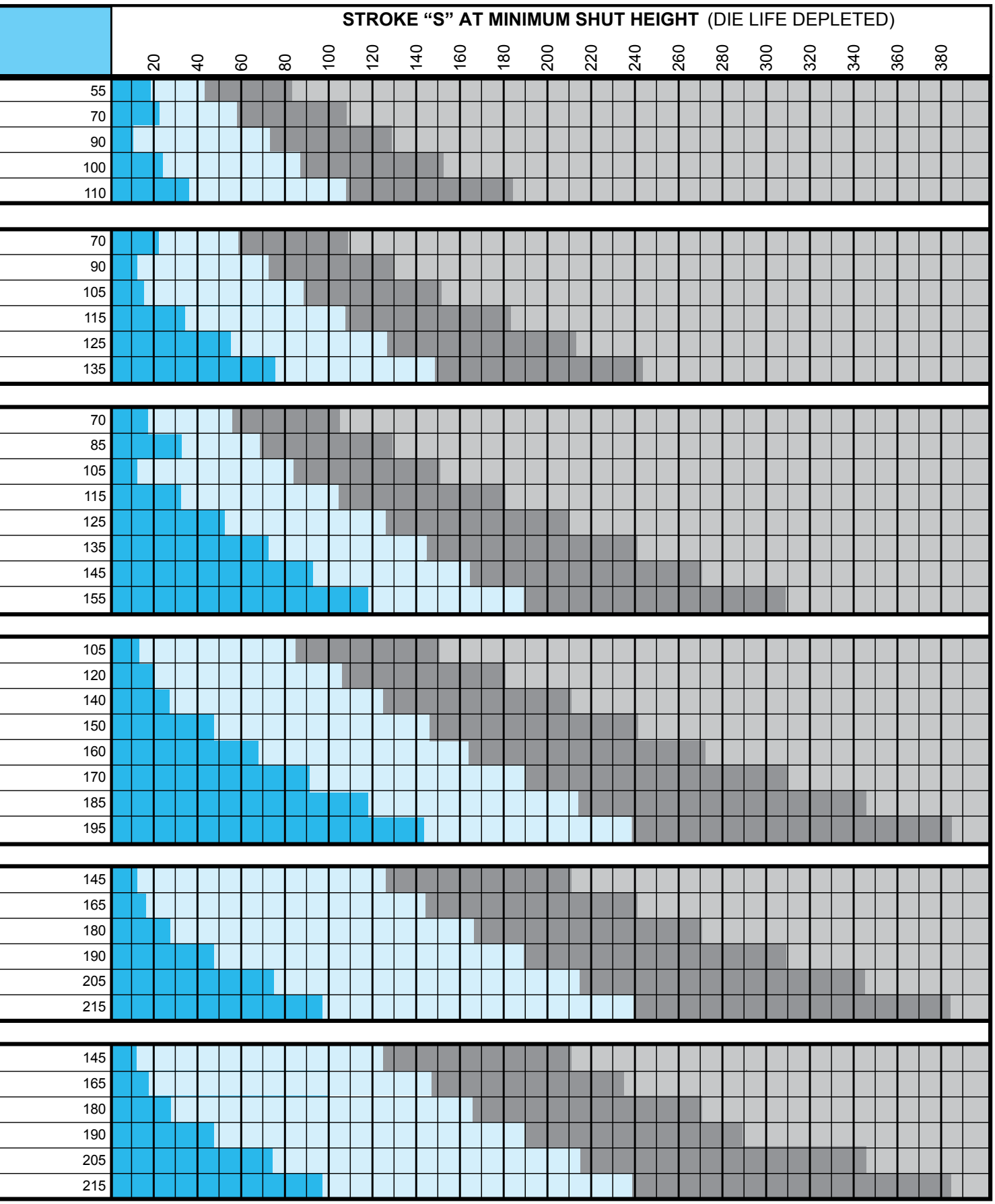
Figure C
Unlimited stroke cage leaves bushing

NOTES:

- Sleeve Bushing: L_2 + K must be less than T
- Shoulder Bushing: J + F + K must be less than T
- Demountable Bushing: Maximum F = T - J - K
- Sleeve Bushing: Maximum L_2 = T - K

Nominal Post Diameter D mm	Ball Bushing			Ball Cage	
	F mm	L1 mm	L2 mm	C mm	
25	35	65	65	55	
	50	80	80	70	
	65	95	95	90	
	–	–	110	100	
	–	–	130	110	
32	50	80	80	70	
	65	95	95	90	
	80	110	110	105	
	–	–	130	115	
	–	–	150	125	
	–	–	170	135	
40	50	80	80	70	
	65	95	95	85	
	80	110	110	105	
	100	130	130	115	
	–	–	150	125	
	–	–	170	135	
	–	–	190	145	
	–	–	215	155	
50	60	110	110	105	
	80	130	130	120	
	100	150	150	140	
	120	170	170	150	
	–	–	190	160	
	–	–	215	170	
	–	–	240	185	
	–	–	265	195	
63	100	150	150	145	
	120	170	170	165	
	140	190	190	180	
	–	–	215	190	
	–	–	240	205	
	–	–	265	215	
80	–	–	150	145	
	–	–	170	165	
	–	–	190	180	
	–	–	215	190	
	–	–	240	205	
	–	–	265	215	

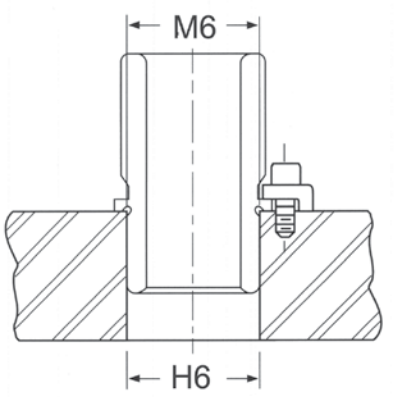
Type II & III
Bushing & Ball Cage Selection Guide



Bore Size Data for Bushings

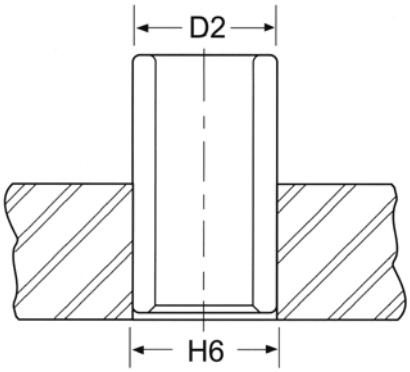
BORE SIZES FOR DEMOUNTABLE PLAIN BEARING BUSHINGS

Bushing Diameter mm	Bushing Locating Diameter M6 mm		Punch Holder Bore H6 mm		Assembly Fit mm	
18 - 19	28	+0.021	28	+0.013	0.005	LOOSE
		+0.008		-0.000	0.021	TIGHT
24 - 25	38	+0.025	38	+0.016	0.007	LOOSE
		+0.009		-0.000	0.025	TIGHT
30 - 32	45	+0.025	45	+0.016	0.007	LOOSE
		+0.009		-0.000	0.025	TIGHT
38 - 40	54	+0.030	54	+0.019	0.008	LOOSE
		+0.011		-0.000	0.030	TIGHT
48 - 50	65	+0.030	65	+0.019	0.008	LOOSE
		+0.011		-0.000	0.030	TIGHT
63	81	+0.035	81	+0.022	0.009	LOOSE
		+0.013		-0.000	0.035	TIGHT
80	100	+0.035	100	+0.022	0.009	LOOSE
		+0.013		-0.000	0.035	TIGHT



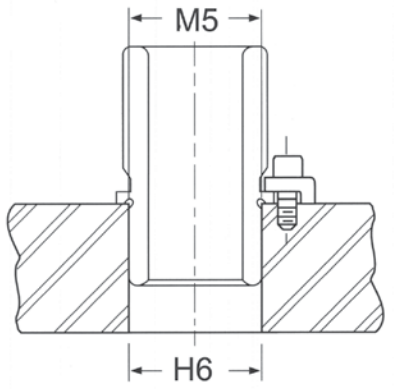
BORE SIZES FOR STRAIGHT SLEEVE BALL BEARING BUSHINGS

Post Diameter mm	Bushing Diameter D2 mm		Die Shoe Bore H6 mm		Assembly Fit mm	
25	45	-0.017	45	+0.016	0.017	LOOSE
		-0.024		-0.000	0.040	
32	54	-0.020	54	+0.019	0.020	LOOSE
		-0.028		-0.000	0.047	
40	65	-0.020	65	+0.019	0.020	LOOSE
		-0.028		-0.000	0.047	
50	81	-0.024	81	+0.022	0.024	LOOSE
		-0.034		-0.000	0.056	
63	95	-0.024	95	+0.022	0.024	LOOSE
		-0.034		-0.000	0.056	
80	112	-0.024	112	+0.022	0.024	LOOSE
		-0.034		-0.000	0.056	



BORE SIZES FOR DEMOUNTABLE BALL BEARING BUSHINGS

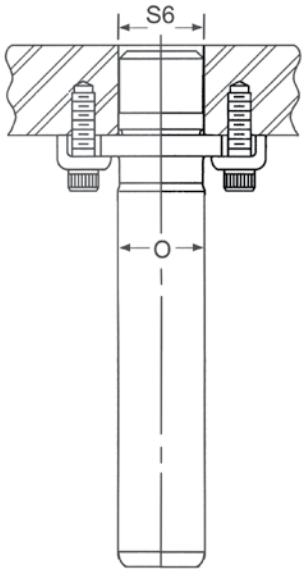
Bushing Diameter mm	Bushing Locating Diameter D2 mm		Die Shoe Bore H6 mm		Assembly Fit mm	
25	45	+0.020	45	+0.016	0.007	LOOSE
		+0.009		-0.000	0.020	TIGHT
32	54	+0.024	54	+0.019	0.008	LOOSE
		+0.011		-0.000	0.024	TIGHT
40	65	+0.024	65	+0.019	0.008	LOOSE
		+0.011		-0.000	0.024	TIGHT
50	81	+0.028	81	+0.022	0.009	LOOSE
		+0.013		-0.000	0.028	TIGHT
63	95	+0.028	95	+0.022	0.009	LOOSE
		+0.013		-0.000	0.028	TIGHT
80	108	+0.032	108	+0.025	0.010	LOOSE
		+0.015		-0.000	0.032	TIGHT



Bore Size Data for Guide Posts

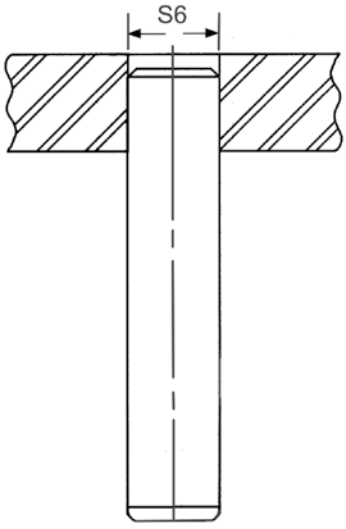
BORE SIZES FOR DEMOUNTABLE GUIDE POSTS

Guide Post Diameter mm		Bore Diameter S6 mm	Assembly Fit mm	
18	-0.016	-0.025	0.002	LOOSE
	-0.027	-0.036	0.020	TIGHT
19	-0.020	-0.031	0.002	LOOSE
	-0.033	-0.044	0.024	TIGHT
24	-0.020	-0.031	0.002	LOOSE
	-0.033	-0.044	0.024	TIGHT
25	-0.020	-0.031	0.002	LOOSE
	-0.033	-0.044	0.024	TIGHT
30	-0.020	-0.031	0.002	LOOSE
	-0.033	-0.044	0.024	TIGHT
32	-0.025	-0.038	0.003	LOOSE
	-0.041	-0.054	0.029	TIGHT
38	-0.025	-0.038	0.003	LOOSE
	-0.041	-0.054	0.029	TIGHT
40	-0.025	-0.038	0.003	LOOSE
	-0.041	-0.054	0.029	TIGHT
48	-0.025	-0.038	0.003	LOOSE
	-0.041	-0.054	0.029	TIGHT
50	-0.025	-0.038	0.003	LOOSE
	-0.041	-0.054	0.029	TIGHT
63	-0.030	-0.047	0.002	LOOSE
	-0.049	-0.066	0.036	TIGHT
80	-0.030	-0.053	0.004	TIGHT
	-0.049	-0.072	0.042	TIGHT



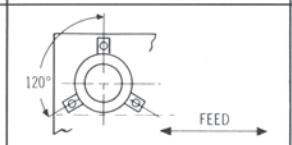
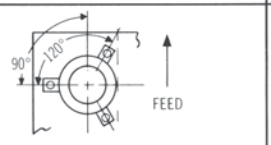
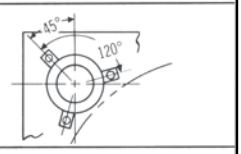
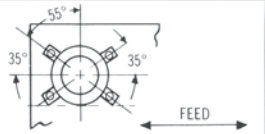
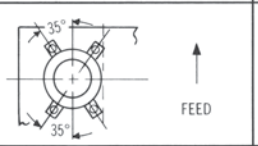
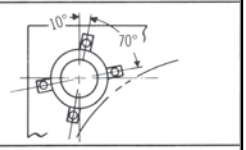
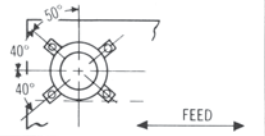
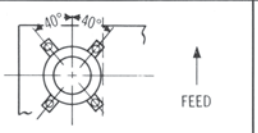
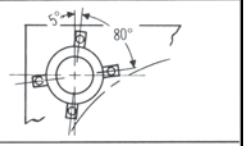
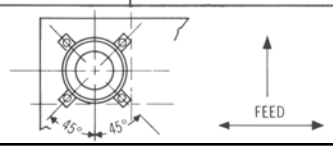
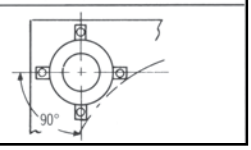
BORE SIZES FOR PRESS FIT GUIDE POSTS

Guide Post Diameter mm	Bore Diameter S6 mm
18	-0.025
	-0.036
19	-0.031
	-0.044
24	-0.031
	-0.044
25	-0.031
	-0.044
30	-0.031
	-0.044
32	-0.038
	-0.054
38	-0.038
	-0.054
40	-0.038
	-0.054
48	-0.038
	-0.054
50	-0.038
	-0.054
63	-0.047
	-0.066
80	-0.053
	-0.072



Clamping Specifications

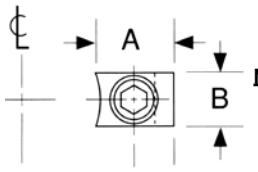
DEMOUNTABLE SHOULDER BUSHING CLAMP ARRANGEMENT SELECTION DATA

NOM. POST DIA.	A (L-R FEED)	B (F-B FEED) (CENTER POST SETS)	C (ROUND DIES)
25 & 32			
40			
50			
63			

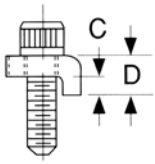
***NOTE:** If not specified, clamp arrangement **A** will be furnished. Clamp arrangement **B** is furnished on Center Post Sets.*

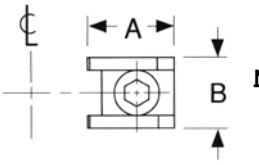
CLAMP ARRANGEMENTS & DIMENSIONS FOR BUSHINGS & GUIDE POSTS

Clamp	Screw	A mm	B mm	C mm	D mm
6-90-1	M6	14.3	15.9	4.9	8.7
6-91-1	M8	19.8	15.9	4.9	9.5
6-93-1	M8	19.6	15.9	6.3	10.9
6-95-1	M6	12.3	12.7	3.2	5.6
6-96-1	M6	17.5	14.5	5	10
6-97-1	M8	24.6	18.9	7.9	13

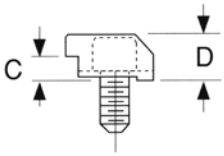


CLAMP PART NUMBERS
6-90-1
6-91-1
6-93-1
6-95-1



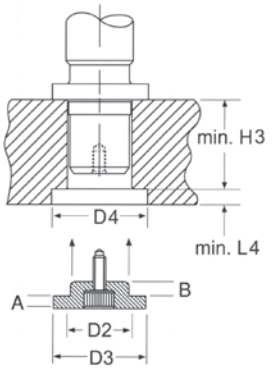


CLAMP PART NUMBERS
6-96-1
6-97-1



RETAINER PLUG DIMENSIONAL DATA

Part Number	D2 mm	D3 mm	A mm	B mm	D4 mm	L4 mm	H3 mm
5-1800-88	18	25	3.3	4.0	26	3.5	25
5-2400-88	24	32	5.0	4.2	33	5.5	30
5-3000-88	30	40	5.0	7.0	41	5.5	40
5-3800-88	38	50	5.0	10.0	51	5.5	51
5-4800-88	48	63	5.0	13.0	64	5.5	59
5-6300-88	63	76	5.0	19.0	77	5.5	70
5-8000-88	80	93	5.0	25.0	95	5.5	87



We offer two ways of mounting the pins into the die plate: clamps or a retainer plug.

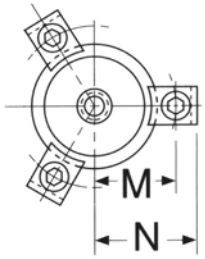
NOTES:

- All tapped holes for retainer plugs are tapped to two (2) times the diameter of the screw.

Clamping Specifications

CLAMP ARRANGEMENTS FOR DEMOUNTABLE BUSHINGS

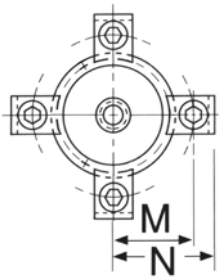
Post Diam. mm	Clamp Part #	Screw Part #	Screw Desc.	Clamps & Screws per Bushing	M mm	N mm
18 & 19	6-95-1	9-06020-91	M6	2	19.5	25.5
24 & 25	6-90-1	9-06020-91	M6	2	27.5	35
30 & 32	6-91-1	9-08020-91	M8 x 20	3	32.5	42.5
38 & 40	6-91-1	9-08020-91	M8 x 20	3	37	47
48 & 50	6-91-1	9-08020-91	M8 x 20	4	44.5	54.5
63	6-91-1	9-08020-91	M8 x 20	4	52	62
80	6-91-1	9-08020-91	M8 x 20	4	62	72



Bushing 3 Clamp Arrangement

CLAMP ARRANGEMENTS FOR LOW PROFILE DEMOUNTABLE BUSHINGS

Post Diam. mm	Clamp Part #	Screw Part #	Screw Desc.	Clamps & Screws per Bushing	M mm	N mm
18 & 19	6-95-1	9-06020-91	M6	2	19.5	25.5
24 & 25	6-96-1	9-06020-91	M6	3	30	37.1
30 & 32	6-96-1	9-06020-91	M6	3	34	41.1
38 & 40	6-97-1	9-08020-91	M8 x 20	4	41	52.1
48 & 50	6-97-1	9-08020-91	M8 x 20	4	47	58.1
63	6-97-1	9-08020-91	M8 x 20	4	56	67.1
80	6-97-1	9-08020-91	M8 x 20	4	67	78.1

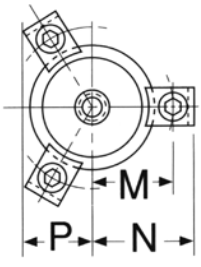


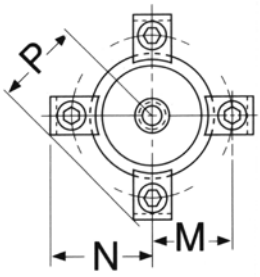
Bushing 2 & 4 Clamp Arrangement

CLAMP ARRANGEMENTS FOR DEMOUNTABLE GUIDE POSTS

Post Diam. mm	Clamp Part #	Screw Part #	Screw Desc.	Clamps & Screws per Post	M mm	N mm	P mm
18 & 19	6-95-1	9-06020-91	M6	3	16.0	22.0	16.5
24 & 25	6-90-1	9-06020-91	M6	3	20.3	27.8	20.8
30 & 32	6-93-1	9-08020-91	M8 x 20	3	25.5	35.5	24.6
38 & 40	6-93-1	9-08020-91	M8 x 20	4	30.0	40.0	26.9
48 & 50	6-93-1	9-08020-91	M8 x 20	4	36.5	46.5	38.5
63	6-93-1	9-08020-91	M8 x 20	4	42.5	52.5	42.7
80	6-93-1	9-08020-91	M8 x 20	4	51.0	61.0	48.8

Guide Post 3 Clamp Arrangement





Guide Post 4 Clamp Arrangement

Component Lubrication

BALL BEARING COMPONENTS

Recommended Lubrication:
- Ball-Lube
- Pint BALL-LUBE spray # ARL0161
- Gallon BALL-LUBE # ARL1281

Notes*:
- Lubricates assemblies providing protection against wear, oxidation and rust
- MSDS available upon request
- Alternative / refined mineral oil of viscosity 290/340 SSU @ 100 degrees F containing “EP” additives and rust inhibitors

SELF-LUBRICATING STYLE BUSHINGS OR WEAR PLATES WITH OIL IMPREGNATED GRAPHITE PLUGS

Recommended Lubrication:
- Light 20 wt. oil should be applied to pre-lube the wear surface of the bushing

Notes*:
- When bushing reaches 80-90 degrees F° because of friction between the components, oil is drawn from the plug thus lubricating the wear surface



NEVER USE GREASE with the Self-Lubricating oil impregnated graphite plug products. Grease on the oil impregnated graphite plugs will prevent the self-lubricating process

PLAIN BEARING FRICTION

Recommended Lubrication:
- Quart # 9-01-52
- Gallon # 9-02-52
- (15) Gallons # 9-02-522
- (55) Gallons # 9-02-521

Notes*:
- Above die lubricant is specially prepared to provide efficient lubrication for guide post in plain bearing applications
- Alternative, lithium complex white grease or multi-purpose grease

***Note:**
MSDS available upon request
Recommendations are based on applications in typical ambient temperatures. For high / low temperature applications, a reputable Lubricant Company or lubrication specialist should be consulted.

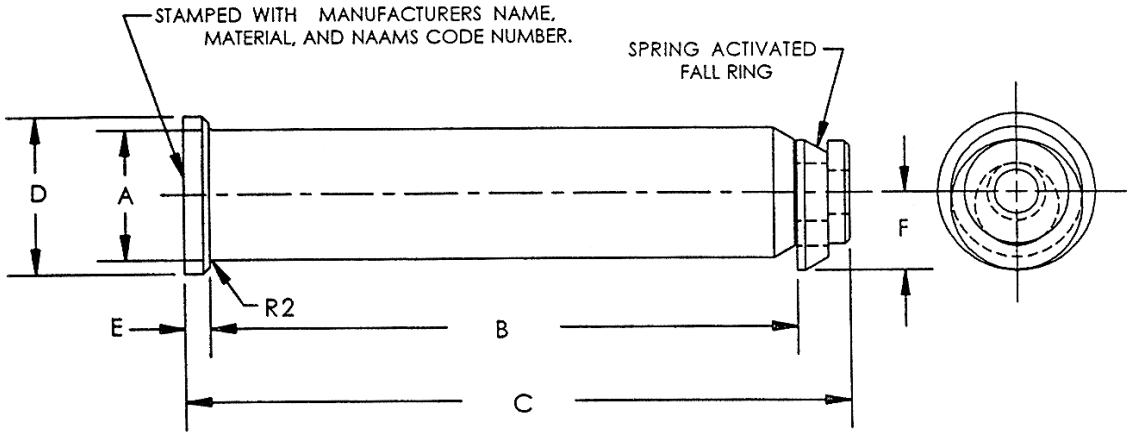
Lifter Pins

Product Features

Lifter pins are manufactured from 1144 steel and are commonly used to handle large die sets. They conform to NAAMS standards.

The internal spring provides added assurance that the fall ring will function properly during the course of the pin's use.





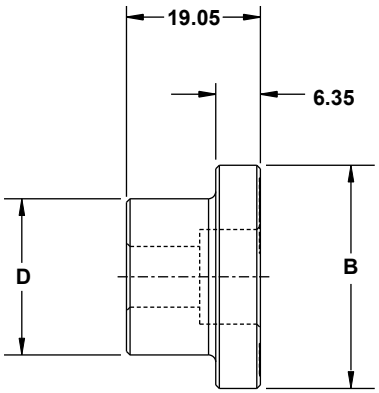
Part Number	NAAMS Code	A $^{+0}_{-.03}$ mm	B mm	C mm	D mm	E mm	F mm	Total Die Wt. in Metric Tons (t)
LP35-125	L013512	35	125	165	45	10	21.5	0 – 2.72
LP50-190	L015019	50	190	230	63	10	31	2.72 – 9.07
LP63-280	L016328	63	280	320	76	10	37.5	9.07 – 36.28
LP80-320	L018032	80	320	370	89	15	46	36.28 – 72.57

Cage Stopper End Cap – Metric

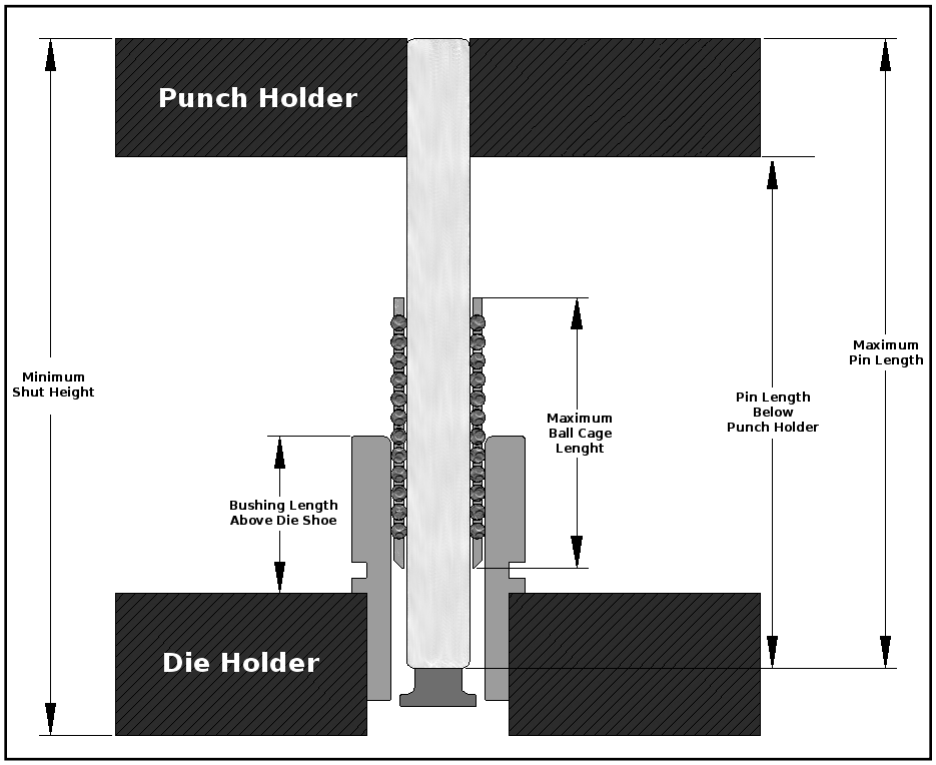


Product Features

The Cage Stopper End Cap is an alternative method to attach the ball bearing cage to a Danly ball bearing guide post. The Cage Stopper attaches to the guide post, allowing the ball bearing cage to rotate when not in preload position. The Cage Stopper is a good option when cage travel required is more than what is available with our traditional Danly ball bearing cages. A socket head cap screw is also included.



Guide Pin Diameter (mm)	B (mm)	D (mm)	SHCS Used	Part Number
25	31.75	22.23	M6	CS-2512
32	38.10	28.58		CS-3212
40	44.45	34.93	M10	CS-4012
50	60.33	47.63		CS-5012
63	73.03	60.33		CS-6312
80	85.73	73.03		CS-8012



Socket Head Cap Screw included

Pad Retainers –
Standard Mount – Metric

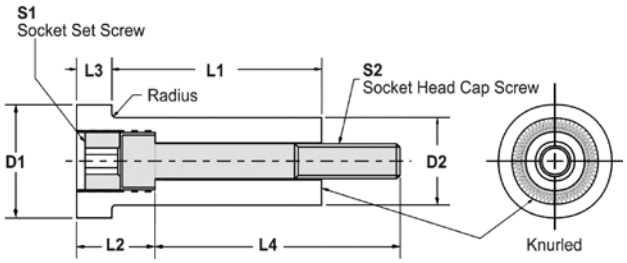
Product Features

These pad retainers are manufactured from 1144 steel and hardened to 28-34 Rockwell C-scale. They are machined to precise tolerances in order to retain the pad and assure parallelism during use.



PART NUMBER	D1	D2	L2	L3	S1	S2	LOAD RATING
NR50	50	40	35	15	M27 x 2	M16 x 2.0	192kN
NR70	70	60	40	15	M36 x 3	M20 x 2.5	299kN

PART NUMBER	Diam. D2 (mm)	Length L1 (mm)	Diam. D1 (mm)	Length L4 (mm)
NR50-30	40	30	50	40
NR50-40		40		45
NR50-45		45		50
NR50-50		50		55
NR50-60		60		65
NR50-65		65		70
NR50-70		70		75
NR50-75		75		75
NR50-90		90		100
NR50-100		100		110
NR50-115		115		120
NR50-130		130		140
NR50-140		140		150
NR50-160		160		170
NR70-40	60	40	70	45
NR70-45		45		50
NR70-50		50		55
NR70-60		60		65
NR70-65		65		70
NR70-70		70		75
NR70-75		75		75
NR70-90		90		100
NR70-100		100		110
NR70-115		115		120
NR70-130		130		140
NR70-140		140		150
NR70-160		160		170

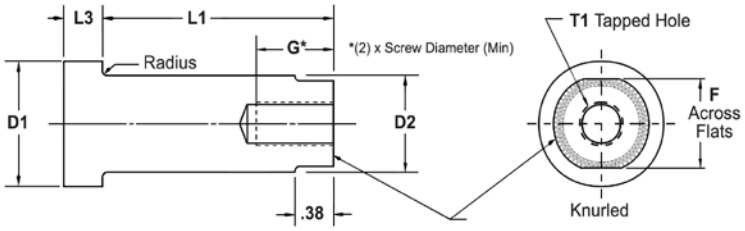


Pad Retainers –
Reverse Mount – Metric



Product Features

These pad retainers are manufactured from 1144 steel and hardened to 28-34 Rockwell C-scale. They are machined to precise tolerances in order to retain the pad and assure parallelism during use.



PART NUMBER	D1	D2	F	L3	LOAD RATING
NR50	50	40	33.65	15	192kN
NR70	70	60	53.70	15	299kN

PART NUMBER	Diam. D2 (mm)	Length L1 (mm)	Diam. D1 (mm)	T1 (mm)
NR50-30R	40	30	50	M16 x 2.0
NR50-40R		40		
NR50-45R		45		
NR50-50R		50		
NR50-60R		60		
NR50-65R		65		
NR50-70R		70		
NR50-75R		75		
NR50-90R		90		
NR50-100R		100		
NR50-115R		115		
NR50-130R		130		
NR50-140R		140		
NR50-160R		160		
NR70-40R	60	40	70	M20 x 2.5
NR70-45R		45		
NR70-50R		50		
NR70-60R		60		
NR70-65R		65		
NR70-70R		70		
NR70-75R		75		
NR70-90R		90		
NR70-100R		100		
NR70-115R		115		
NR70-130R		130		
NR70-140R		140		
NR70-160R		160		

PINS & BUSHINGS – METRIC

The Danly IEM Value Proposition

I. **DANLY IEM** is recognized as the leader in manufacturing quality die components to the global parts forming industry. Our reputation has been built by satisfying customer needs, and we are very strong in the automotive and appliance industries.

II. **DANLY IEM** offers outstanding delivery on a consistent basis. Choosing us as a supplier means that our customers have a competitive advantage in delivering their products to the market.

III. **DANLY IEM** has complex machining capabilities on die components at several facilities. With extensive machining capabilities in the USA and China, we have taken the lead role in creating and bringing new products to customers and helping them find solutions that improve their operations.

IV. **DANLY IEM's** vast breadth of products assures innovative solutions. We strive to address customer problems by utilizing our research and development department as well as other technical professionals.

V. **DANLY IEM** has a technically trained sales force and distributor channels with Engineering support. Sales, Marketing and Engineering professionals are available to support our product lines.

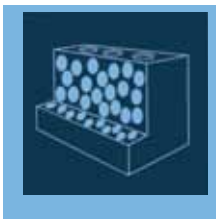
- **Competitive Prices**
- **Reliability and Performance**



...A LEADING MANUFACTURER AND INNOVATOR OF DIE COMPONENTS SUPPLIED GLOBALLY TO THE METAL FORMING INDUSTRY...









- **High Quality Design & Construction**
- **Outstanding Service & Support**

Commitment to Quality & Customer Satisfaction

Dayton Lamina is a leading manufacturer of tool, die and mold components for the metal-working and plastics industries. As a customer-focused, world-class supplier of choice, we provide the brands, product breadth, distribution network and technical support for all your metal forming needs.

Our goal is to give our customers the most innovative and value-added products and services.



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